

CHAPTER 11

THERMAL PROPERTIES & TEMPERATURE

2.2 Thermal properties and temperature

2.2.1 Thermal expansion of solids, liquids and gases

- 1 Explain applications and consequences of thermal expansion in the context of common examples, including the liquid-in-glass thermometer
- 2 Explain, in terms of the motion and arrangement of particles, the thermal expansion of solids, liquids and gases, and state the relative order of magnitudes of the expansion of solids, liquids and gases
- 3 Convert temperatures between kelvin and degrees Celsius; recall and use the equation
 $T \text{ (in K)} = \theta \text{ (in } ^\circ\text{C)} + 273$

2.2.2 Specific heat capacity

- 1 Know that an increase in the temperature of an object increases its internal energy
- 2 Describe an increase in temperature of an object in terms of an increase in the average kinetic energies of all of the particles in the object
- 3 Define specific heat capacity as the energy required per unit mass per unit temperature increase; recall and use the equation

$$\text{specific heat capacity} = \frac{\text{change in energy}}{\text{mass} \times \text{change in temperature}}$$

$$c = \frac{\Delta E}{m\Delta\theta}$$

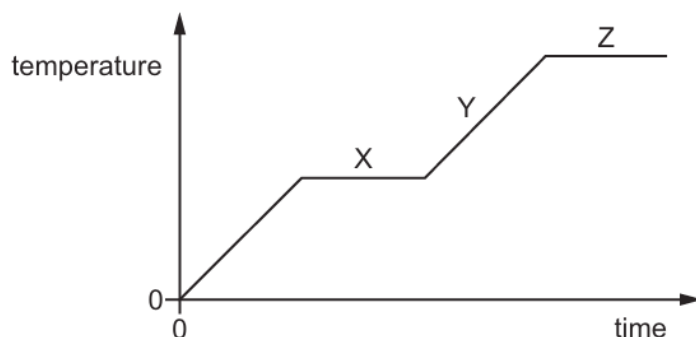
- 4 Describe experiments to measure the specific heat capacity of a solid and of a liquid

2.2.3 Melting, boiling and evaporation

- 1 Describe melting, solidification, boiling and condensation in terms of energy transfer without a change in temperature
- 2 Know the melting and boiling temperatures for water at standard atmospheric pressure
- 3 Describe the differences between boiling and evaporation
- 4 Describe evaporation in terms of the escape of more energetic particles from the surface of a liquid
- 5 Describe how temperature, surface area and air movement over a surface affect evaporation
- 6 Explain how evaporation causes cooling
- 7 Describe latent heat as the energy required to change the state of a substance and explain it in terms of particle behaviour and the forces between particles

1. O/N/23/11/Q15

A graph of temperature against time is shown for a material being heated from its solid state.

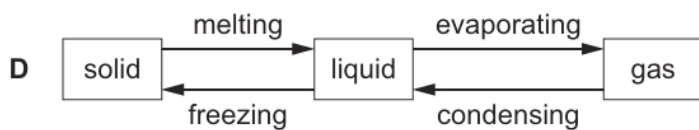
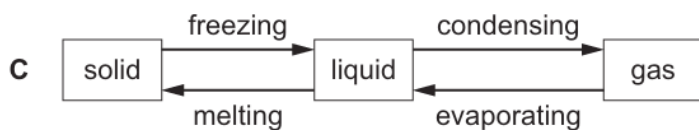
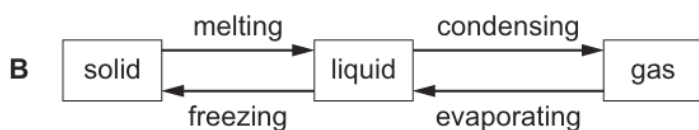
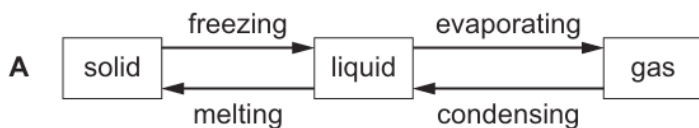


Which row describes what is happening at X, Y and Z?

	X	Y	Z
A	boiling	solid becomes hotter	melting
B	liquid becomes hotter	boiling	gas becomes hotter
C	solid becomes hotter	melting	liquid becomes hotter
D	melting	liquid becomes hotter	boiling

2. O/N/23/12/Q17

Which diagram shows the names of the changes of state that correspond to the directions of the arrows?



3. O/N/23/12/Q18

A sealed cylinder containing gas is left in direct sunlight.

Eventually the cylinder explodes.

How is this explained in terms of the behaviour of the gas particles?

- A** Each particle gets bigger and they collide with each other more often.
- B** Each particle gets bigger and they collide harder with the cylinder.
- C** The particles gain kinetic energy and hit the cylinder harder and more often.
- D** The particles lose kinetic energy and they hit the cylinder harder and more often.

4. O/N/23/12/Q20

Pot X and pot Y are both in a cool room.

Pot X contains hot water below its boiling point.

Pot Y is identical to pot X but contains water that is kept boiling by a hotplate underneath it.



What happens to the average kinetic energy of the particles in the liquid water in pot X and to the average kinetic energy of the particles in the liquid water in pot Y?

	kinetic energy of liquid particles in X	kinetic energy of liquid particles in Y
A	decreases	increases
B	decreases	does not change
C	stays the same	increases
D	stays the same	does not change

5. M/J/23/11/Q16

An incomplete statement about latent heat is given.

Latent heat is the1..... required to overcome the bonds between molecules and change the2..... of a substance.

Which words are inserted into gap 1 and gap 2 to correctly complete the sentence?

	gap 1	gap 2
A	energy	state
B	force	state
C	energy	temperature
D	force	temperature

6. M/J/23/11/Q17

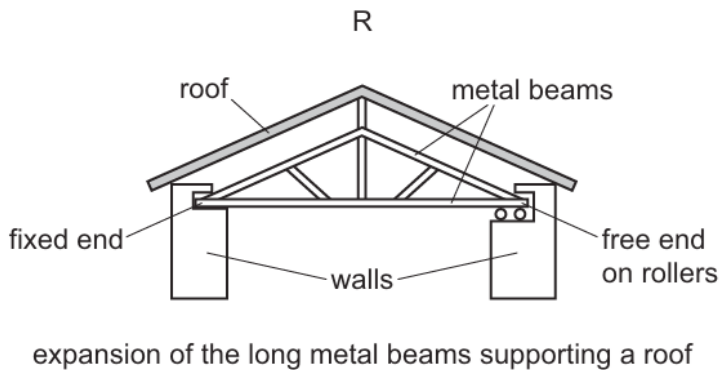
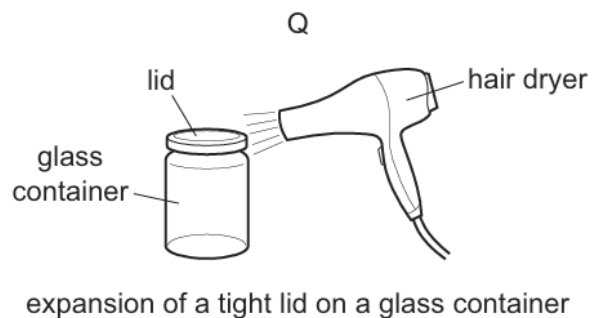
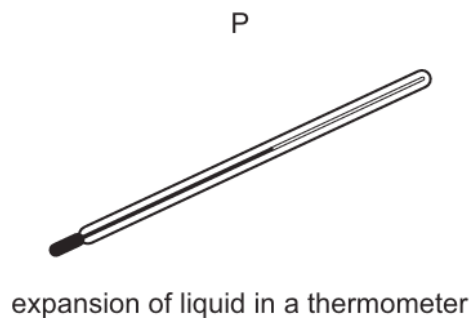
A cold solid is placed on top of a hot solid. Thermal energy is transferred from the hot solid to the cold one.

What is the explanation for this?

- A** A hot solid expands, so its particles will move further apart.
- B** Energy is transferred from one particle to the next.
- C** Heat always rises.
- D** Molecules are free to move randomly through the solids.

7. M/J/23/12/Q16

The diagrams show examples of thermal expansion.



In which diagrams is the thermal expansion useful?

- A** P, Q and R **B** P and Q only **C** P only **D** Q and R only

8. O/N/22/11/Q20

A piece of iron of mass m is placed in a mixture of ice and water.

Its temperature decreases from θ_1 to θ_2 .

How much thermal energy is lost by the piece of iron?

- A** $m \times \text{specific heat capacity of iron} \times (\theta_1 - \theta_2)$
- B** $m \times \text{specific heat capacity of iron} \times \theta_2$
- C** $m \times \text{heat capacity of piece of iron} \times (\theta_1 - \theta_2)$
- D** $m \times \text{heat capacity of piece of iron} \times \theta_2$

9. O/N/22/11/Q21

A kilogram of a substance has a fixed shape.

Thermal energy supplied to the substance weakens the forces of attraction between its molecules.

What is happening?

- A** A gas is being heated.
- B** A liquid is at its boiling point.
- C** A liquid is evaporating.
- D** A solid is at its melting point.

10. O/N/22/11/Q22

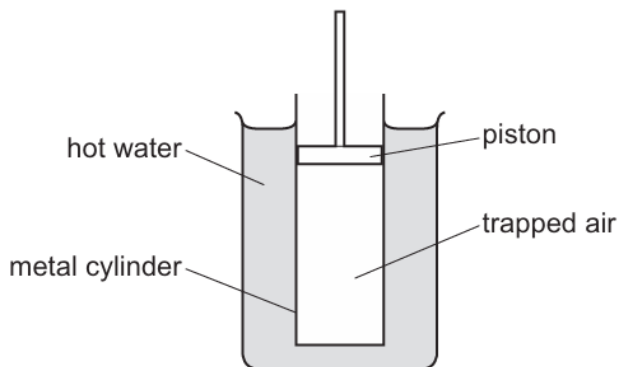
Which conditions produce the greatest rate of evaporation from a liquid?

- A** high temperature, draught and large surface area of liquid
- B** high temperature, no draught and large surface area of liquid
- C** low temperature, draught and small surface area of liquid
- D** low temperature, no draught and small surface area of liquid

11. O/N/22/12/Q21

Air is trapped in a metal cylinder by a piston. The piston is free to move and the trapped air is at atmospheric pressure.

The cylinder is in hot water.



The cylinder is taken out of the water and left to cool.

What happens to the mass of the air in the cylinder and its pressure as it cools?

	mass of air	pressure of air
A	does not change	does not change
B	does not change	changes
C	changes	does not change
D	changes	changes

12. M/J/22/11/Q17

What is the usual symbol and what is the unit for the heat capacity of an object?

	symbol	unit
A	C	$\text{J}/^{\circ}\text{C}$
B	C	$\text{J}/(\text{kg } ^{\circ}\text{C})$
C	H	$\text{J}/^{\circ}\text{C}$
D	H	$\text{J}/(\text{kg } ^{\circ}\text{C})$

13. M/J/22/12/Q20

Steam at 110°C condenses on a surface to form water droplets at 100°C .

What happens after the steam comes into contact with the surface?

- A** The molecules slow down and absorb energy from the surroundings.
- B** The molecules slow down and emit energy to the surroundings.
- C** The molecules stay at the same speed and absorb energy from the surroundings.
- D** The molecules stay at the same speed and emit energy to the surroundings.

14. M/J/22/12/Q21

Which list shows the states of matter in order of expansion from smallest to greatest for the same temperature rise?

- A** gas → liquid → solid
- B** liquid → solid → gas
- C** solid → gas → liquid
- D** solid → liquid → gas

15. M/J/22/12/Q22

Which row describes the boiling and the evaporation of a liquid?

	boiling	evaporation
A	bubbles form throughout the liquid	only occurs at one temperature
B	bubbles form throughout the liquid	produces cooling
C	occurs at any temperature	only occurs at one temperature
D	occurs at any temperature	produces cooling

16. M/J/22/12/Q23

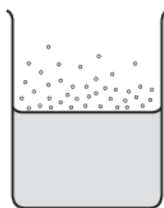
A hairdryer is used to blow air across the surface of water in a beaker so that the water evaporates.

What increases the rate of evaporation of the water?

- A** decreasing the speed of the air from the hairdryer
- B** decreasing the mass of the water in the beaker
- C** increasing the surface area of the water by using a wider beaker
- D** increasing the volume of the water in the beaker

17. O/N/21/11/Q16

The diagram shows a room temperature liquid in a beaker.



What reduces the rate of loss of liquid by evaporation?

- A** blowing air across the top of the beaker
- B** heating the liquid
- C** putting a lid on the beaker
- D** transferring the liquid to a beaker with a larger diameter

18. O/N/21/12/Q18

Solar panels are used to heat water with a mass of 500 kg.

The total area of the solar panels is 10 m^2 and the average power output from each square metre of the panels is 300 W. The specific heat capacity of water is $4200 \text{ J/(kg } ^\circ\text{C)}$.

What is the increase in the water temperature after 8.0 hours? (Assume there is no heat loss.)

- A** 0.69°C **B** 4.1°C **C** 5.1°C **D** 41°C

19. O/N/21/12/Q20

The temperatures of the water in four beakers are different and areas of the surfaces of the water are also different.

In which beaker is the rate of evaporation of the water greatest?

	temperature of water / $^\circ\text{C}$	surface area of water / cm^2
A	20	50
B	20	100
C	25	50
D	25	100

20. M/J/21/11/Q18

What is the *heat capacity* of a body?

- A** the amount of thermal energy that the body can absorb without melting
- B** the amount of thermal energy required to raise the temperature of the body by 1.0°C
- C** the amount of thermal energy required to raise the temperature of 1.0 kg of the body by 1.0°C
- D** the amount of thermal energy required to raise the temperature of 1.0 m^3 of the body by 1.0°C

21. M/J/21/11/Q19

Which statement about water is correct?

- A** At the boiling point, water vapour molecules have the same kinetic energy as liquid water molecules.
- B** Evaporation occurs only at the boiling point.
- C** Water molecules become heavier when water freezes.
- D** Water molecules lose all of their kinetic energy when water freezes.

22. M/J/21/12/Q21

Which statement about the thermal expansion of solids, liquids and gases is correct?

- A** Liquids do not expand.
- B** Liquids expand more than gases but less than solids.
- C** Liquids expand more than solids but less than gases.
- D** Liquids expand the same amount as solids but less than gases.

23. O/N/20/11/Q21

What is a unit for heat capacity?

- A** J
- B** J/g
- C** J/°C
- D** J/(g °C)

24. O/N/20/11/Q22

A block of aluminium is heated.

What happens to the kinetic energy and to the potential energy of the molecules?

	kinetic energy	potential energy
A	increase	increase
B	increase	decrease
C	decrease	increase
D	decrease	decrease

25. O/N/20/11/Q23

Which statement about the melting of ice and the boiling of water is correct?

- A** Both processes are accompanied by a decrease in volume.
- B** Both processes involve energy transfer and a temperature change.
- C** Both processes involve the absorption of latent heat.
- D** Both processes result in the increase of intermolecular forces.

26. O/N/20/11/Q24

A substance has a melting point of -17°C and a boiling point of 117°C .

In which state does the substance exist at -10°C and at 110°C ?

	at -10°C	at 110°C
A	solid	liquid
B	solid	gas
C	liquid	liquid
D	liquid	gas

27. M/J/20/12/Q24

Thermal energy is used to turn water at 100 °C to steam at 100 °C.

Which of the following occurs?

- 1 The forces between the molecules decrease.
- 2 The speed of the molecules increases.
- 3 The separation of the molecules increases.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

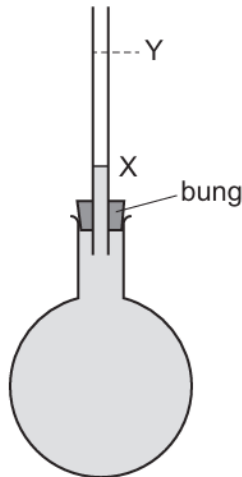
28. O/N/19/11/Q21

What is the name of the change of state that occurs when a substance changes from a solid to a liquid?

- A** boiling
B condensation
C evaporation
D melting

29. O/N/19/11/Q23

The diagram shows a glass flask full of water at 10 °C. A long glass tube passes through the bung into the water. The water level in the tube is at X.



The flask is placed in hot water and the water level in the tube eventually rises to Y.

Why does the water level rise?

- A** The flask contracts and the water expands.
B The flask contracts more than the water contracts.
C The flask expands and the water contracts.
D The flask expands less than the water expands.

30. O/N/19/11/Q24

A glass beaker contains water in a warm room.

The water is initially at the same temperature as the room.

Which change does **not** affect the initial rate of evaporation of the water?

- A** changing the temperature of the water
- B** using a beaker made of thicker glass
- C** using a beaker with a larger cross-sectional area
- D** using a fan to blow air over the beaker

31. O/N/19/12/Q18

A student makes four statements about evaporation and boiling.

- 1 Both involve an increase in molecular separation.
- 2 Both occur only at a fixed temperature.
- 3 Both involve the absorption of thermal energy.
- 4 Both occur throughout the liquid.

Which two statements are correct?

- A** 1 and 4 **B** 2 and 4 **C** 1 and 3 **D** 2 and 3

32. O/N/19/12/Q21

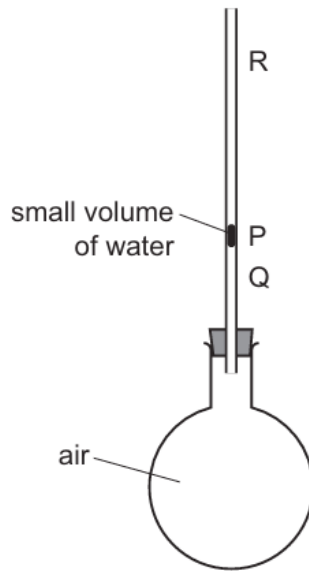
A metal disc is heated to 600 °C and then lowered into a beaker of water.

What happens to the mass of the metal disc and to the mass of the water in the beaker?

	mass of metal disc	mass of water in beaker
A	increases	decreases
B	increases	no change
C	no change	decreases
D	no change	increases

33. M/J/19/11/Q20

The diagram shows air trapped in a flask by a small volume of water in a thin tube.



When the flask is held in a student's hands, the small volume of water first moves down from P to Q, and then up to R.

Why does the small volume of water move like this?

- A The flask contracts and then the air expands.
- B The flask expands and then the air contracts.
- C The flask expands and then the air expands less than the flask.
- D The flask expands and then the air expands more than the flask.

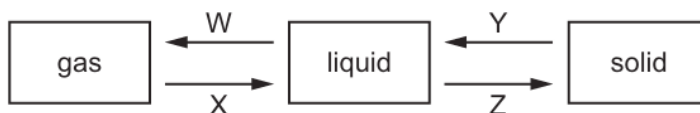
34. M/J/19/12/Q22

How is *latent heat* defined?

- A the thermal energy absorbed or emitted by unit mass during a change of state without a temperature change
- B the thermal energy absorbed or emitted by unit mass during unit temperature change
- C the thermal energy absorbed or emitted during a change of state without a temperature change
- D the thermal energy absorbed or emitted during unit temperature change

35. O/N/18/11/Q13

The diagram shows four changes of state W, X, Y and Z.



Which row gives the names of the four changes of state?

	W	X	Y	Z
A	condensation	boiling	freezing	melting
B	condensation	boiling	melting	freezing
C	boiling	condensation	freezing	melting
D	boiling	condensation	melting	freezing

36. O/N/18/11/Q14

What increases when a liquid becomes a gas at its boiling point?

- A** the average kinetic energy of the molecules
- B** the molecular size
- C** the molecular spacing
- D** the total number of molecules

37. O/N/18/11/Q15

A metal lid fits tightly on a glass jar.

Which process makes it easier to remove the lid from the jar?

- A** cool the lid only
- B** put the jar and lid in a refrigerator
- C** warm the jar only
- D** warm the lid only

38. O/N/18/11/Q16

The specific heat capacity of lead is $130 \text{ J/(kg } ^\circ\text{C)}$. Lead of mass 0.50 kg is heated from 10°C to 35°C .

Which calculation gives the amount of thermal energy, in J, absorbed by the lead?

- A** $0.50 \times 130 \times 25$
- B** $0.50 \times 130 \times 35$
- C** $\frac{0.50 \times 130}{25}$
- D** $\frac{0.50 \times 130}{35}$

39. M/J/18/11/Q17

What is described as 'the escape of more energetic molecules from the surface of a liquid'?

- A condensation
- B evaporation
- C melting
- D radiation

40. M/J/18/11/Q19

What happens when a solid is heated and expands?

- A The molecules do not change size and the spaces between the molecules become larger.
- B The molecules expand and the spaces between the molecules become larger.
- C The molecules expand and the spaces between the molecules become smaller.
- D The molecules expand and the spaces between the molecules do not change size.

41. O/N/17/11/O12

Scientists believe that some lakes are shrinking because of evaporation.

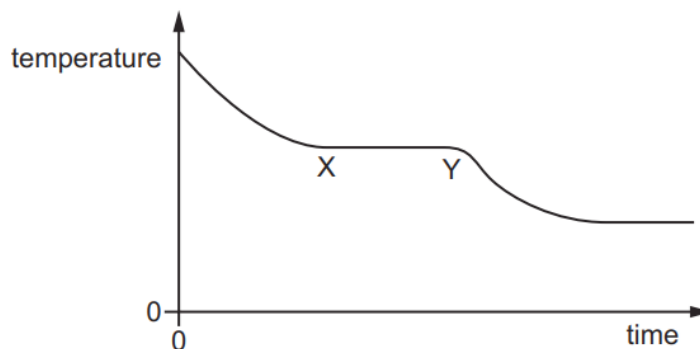
What increases the rate of evaporation?

- A a decrease in the surface area
- B a fall in water temperature
- C an increase in the depth of the lake
- D wind blowing across the surface

42. O/N/17/11/Q13

Solid wax is melted in a boiling tube and then allowed to cool.

The graph shows the cooling curve.



What is happening between points X and Y?

- A The liquid is at its boiling point.
- B The liquid is turning to solid.
- C The solid has reached room temperature.
- D The solid is turning to liquid.

43. O/N/17/11/Q14

During boiling and during melting, a substance either takes in energy or gives out energy.

Which row gives the energy transfer corresponding to the change of state?

	during boiling	during melting
A	gives out energy	gives out energy
B	gives out energy	takes in energy
C	takes in energy	gives out energy
D	takes in energy	takes in energy

44. O/N/17/11/Q15

The temperature of a 50 g mass of a metal is raised by 40 °C. The specific heat capacity of the metal is 0.40 J / (g °C).

How much thermal energy is supplied?

- A** 0.32 J **B** 0.50 J **C** 500 J **D** 800 J

45. O/N/17/12/Q13

A dish of liquid is left on a laboratory bench. Some of the liquid evaporates.

What happens and why?

- A** The liquid cools because liquid molecules have more potential energy than gas molecules.
B The liquid cools because faster-moving molecules escape.
C The liquid warms because liquid molecules have less potential energy than gas molecules.
D The liquid warms because slower-moving molecules are left behind.

46. O/N/17/12/Q15

What is meant by the *heat capacity* of a solid object?

- A** the maximum thermal energy it can store per unit mass
B the maximum thermal energy it can store per unit volume
C the thermal energy required to cause a unit temperature rise
D the thermal energy required to raise its temperature to its melting point

47. M/J/17/11/Q20

A substance consists of particles that are close together and moving past each other at random. The average speed of the particles is gradually increasing.

What best describes the substance?

- A** a gas being heated
- B** a liquid being heated
- C** a liquid undergoing solidification
- D** a solid being heated

48. M/J/17/11/Q21

The table shows the melting points and the boiling points of some elements.

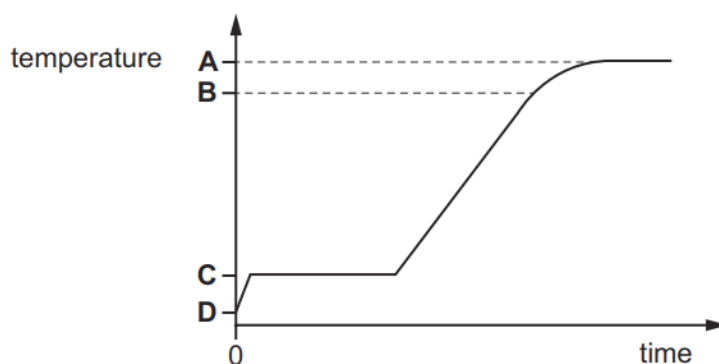
Which element is a liquid over the largest range of temperature?

	element	melting point /°C	boiling point /°C
A	bromine	−7	59
B	chlorine	−102	−34
C	fluorine	−220	−188
D	iodine	114	184

49. M/J/17/11/Q22

Some ice cubes are taken from a freezer and heated in a container. The readings of temperature and time are recorded on the graph.

Which temperature is 0 °C?



50. M/J/17/11/Q25

There is a tight metal lid on a glass jar. The lid is loosened by pouring hot water over it.

Which physics property explains this?

- A** convection
- B** evaporation
- C** expansion
- D** radiation

51. O/N/16/11/Q16

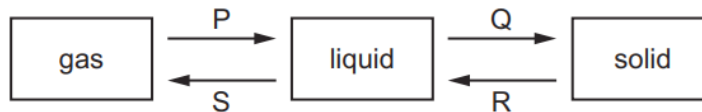
When a liquid evaporates, molecules escape from its surface.

Which molecules escape, and what happens to the average kinetic energy of the molecules remaining in the liquid?

- A** The less energetic molecules escape and the average kinetic energy decreases.
- B** The less energetic molecules escape and the average kinetic energy increases.
- C** The more energetic molecules escape and the average kinetic energy decreases.
- D** The more energetic molecules escape and the average kinetic energy increases.

52. O/N/16/11/Q17

The diagram shows changes of state.

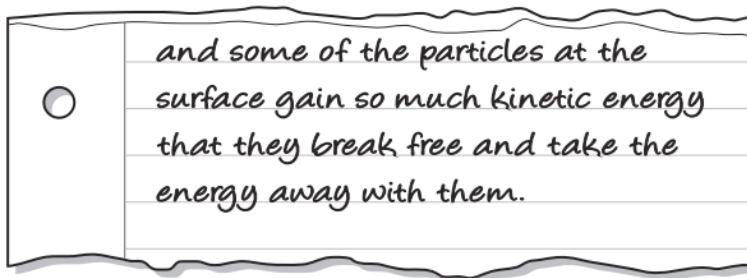


What are the names of the changes of state P, Q, R and S?

	P	Q	R	S
A	condensation	melting	solidification	boiling
B	condensation	solidification	melting	boiling
C	boiling	melting	solidification	condensation
D	boiling	solidification	melting	condensation

53. O/N/16/12/Q19

A piece of paper torn out of an exercise book is shown.



Which process is being described?

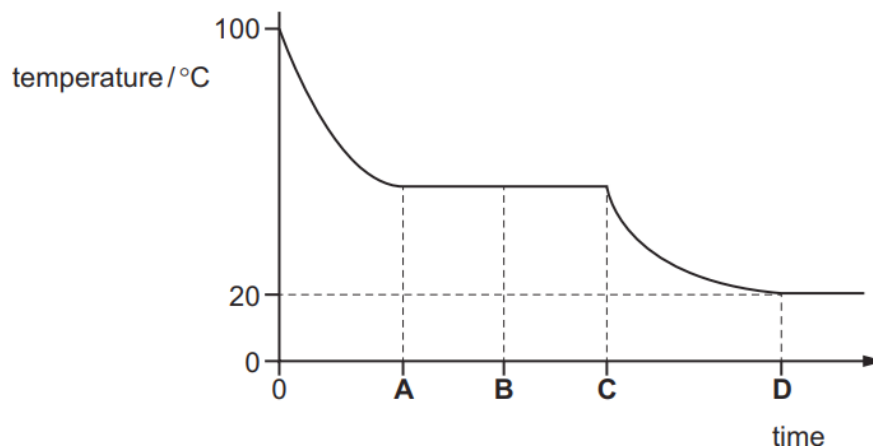
- A conduction
- B convection
- C evaporation
- D radiation

54. O/N/16/12/Q20

A sample of liquid at 100°C is allowed to cool to a room temperature of 20°C .

The diagram shows how the temperature changes with time during the cooling process.

At which time does the sample become completely solid?



55. O/N/16/12/Q21

One kilogram of water needs more thermal energy to raise its temperature by 1°C than one kilogram of copper.

Why is this?

- A copper has a higher specific heat capacity than water
- B copper has a lower specific heat capacity than water
- C copper is a better conductor of heat than water
- D copper is a poorer conductor of heat than water

56. O/N/15/11/Q17

An electric shower takes in cold water at 17°C . The shower gives 6000 J of energy every second to the cold water and heats it to 37°C . The specific heat capacity of water is $4200\text{ J}/(\text{kg }^{\circ}\text{C})$.

What is the mass of hot water supplied by the shower in one second?

- A** 0.035 kg **B** 0.039 kg **C** 0.071 kg **D** 0.084 kg

57. O/N/15/12/Q18

Some of the liquid in a dish evaporates, as shown in the diagrams.



Which molecules leave the liquid and which molecules in the liquid have greater average kinetic energy?

	molecules that leave have	molecules in the liquid have greater average kinetic energy
A	high energy	before evaporation
B	high energy	after evaporation
C	low energy	before evaporation
D	low energy	after evaporation

58. O/N/15/12/Q19

An insulated beaker contains 300 g of water, initially at 30°C . Water at 100°C is added until the temperature of the mixture reaches 50°C .

The specific heat capacity of water is $4.2\text{ J}/(\text{g }^{\circ}\text{C})$.

How much water is added?

- A** 60 g **B** 120 g **C** 180 g **D** 750 g

59. M/J/15/11/Q19

Which substance in the table is liquid at 20°C ?

	melting point $/^{\circ}\text{C}$	boiling point $/^{\circ}\text{C}$
A	-218	-183
B	-39	357
C	44	280
D	119	444

60. M/J/15/12/Q21

In a liquid, some energetic molecules break free from the surface even when the liquid is too cold for bubbles to form.

What is the name of this process?

- A** boiling
- B** condensation
- C** convection
- D** evaporation

61. M/J/15/12/Q22

A pupil adds 37 g of ice at 0 °C to 100 g of water at 30 °C. The final temperature of the water and melted ice is 0 °C. No heat is lost to, or gained from, the surroundings.

The specific heat capacity of water is 4.2 J/(g °C).

What is the specific latent heat of ice?

- A** 47 J/g **B** 341 J/g **C** 4700 J/g **D** 12 600 J/g

62. M/J/15/12/Q23

The heat capacity of an object, of mass 2.0 kg, is C . The energy needed to

- A** increase the temperature of the whole object by Δt is $C\Delta t$.
- B** increase the temperature of unit mass of the object by Δt is $C\Delta t$.
- C** melt the whole object is C .
- D** melt unit mass of the object is C .

63. O/N/14/11/Q18

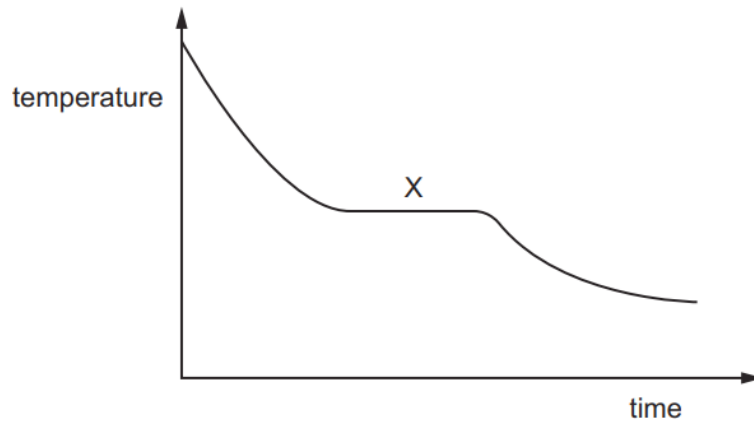
Water is poured into four dishes. In two of the dishes the water has a small surface area and in the other two it has a large surface area. The water in two of the dishes is cool and the water in the other two is warm.

From which dish does the water evaporate the quickest?

	surface area	temperature
A	large	cool
B	large	warm
C	small	cool
D	small	warm

64. M/J/14/11/Q20

A hot liquid is poured into a beaker. The graph shows how the temperature of the liquid changes as it cools towards room temperature.

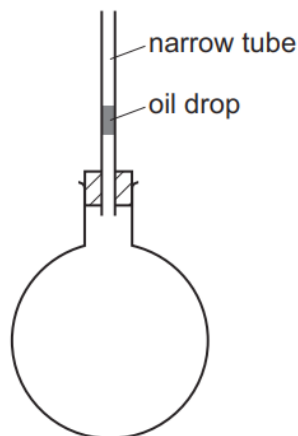


What is occurring at region X?

- A boiling and evaporation
- B condensation only
- C evaporation only
- D solidification and evaporation

65. M/J/14/12/Q21

The diagram shows a flask containing air. The air is trapped by a drop of oil in a narrow tube.



When the flask is heated the oil drop rises up the tube.

Which statement is **not** correct?

- A The air molecules each get larger.
- B The air molecules hit the container with greater force.
- C The air molecules move faster.
- D The air molecules move further apart.

66. M/J/14/12/Q22

Thermal energy is transferred to a solid. First it melts and then it boils to produce a gas.

Which statement about the temperature is correct?

- A** When melting and boiling the temperature does not change.
- B** When melting and boiling the temperature increases.
- C** When melting the temperature increases but when boiling the temperature stays the same.
- D** When melting the temperature stays the same but when boiling the temperature increases.

67. O/N/13/11/Q14

The temperature shown by a mercury-in-glass thermometer increases.

Which of the following is constant?

- A** the density of the mercury
- B** the internal energy of the mercury
- C** the mass of the mercury
- D** the volume of the mercury

68. O/N/13/11/Q17

When a person climbs out of a warm swimming pool on a hot summer day, he feels cold.

Why does this happen?

- A** The air has a high specific heat capacity.
- B** The air is a better conductor of heat than water.
- C** The water droplets increase his surface area.
- D** The water takes heat from his body to evaporate.

69. O/N/13/12/Q18

Different amounts of energy are supplied to copper blocks of different masses.

Which block experiences the greatest temperature change?

	mass of block / kg	energy supplied / J
A	0.1	200
B	0.2	200
C	0.4	600
D	0.8	400

70. O/N/13/12/Q21

The more energetic molecules in a liquid may escape from its top surface.

What is this process called?

- A** boiling
- B** Brownian motion
- C** convection
- D** evaporation

71. M/J/13/12/Q17

A liquid-in-glass thermometer contains mercury.

Which physical property of the mercury varies with temperature, enabling the thermometer to operate?

- A** mass
- B** melting point
- C** resistance
- D** volume

72. M/J/13/12/Q18

Thermal energy of 12 000 J is supplied to a 2.0 kg mass of copper.

The specific heat capacity of copper is 400 J / (kg °C).

What is the rise in temperature?

- A** 15 °C
- B** 30 °C
- C** 60 °C
- D** 100 °C

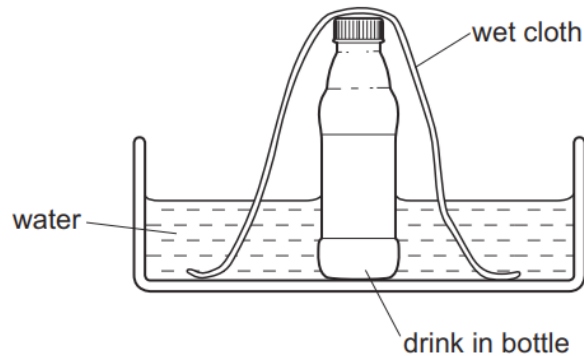
73. M/J/13/12/Q21

What happens to the molecules of a gas when the gas changes into a liquid?

- A** They move closer and lose energy.
- B** They move closer and gain energy.
- C** They move apart and lose energy.
- D** They move apart and gain energy.

74. O/N/12/11/Q16

On a hot day, the drink in a bottle can be kept cool by standing the bottle in a bowl of water and placing a wet cloth over it.

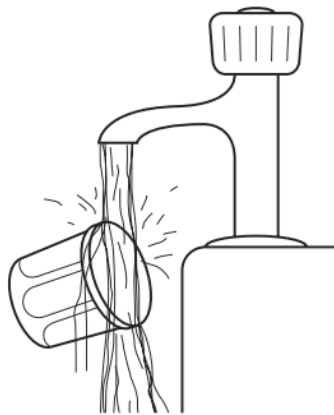


Why is the drink kept cool?

- A** Hot air cannot escape from the bottle.
- B** The cloth conducts heat from the bottle into the water.
- C** The drink cannot evaporate from the bottle.
- D** Water evaporating from the cloth cools the drink.

75. O/N/12/11/Q20

A person cannot unscrew the metal lid of a pot of jam. The lid can be unscrewed after it has been held under hot, running water for a few seconds.



Why is this?

- A** The air pressure in the jar falls.
- B** The glass expands.
- C** The jam melts.
- D** The metal lid expands.

76. O/N/12/12/Q16

In hot weather, people use electric fans to keep cool.

Why do the fans make them feel cool?

- A** They change one form of energy into another.
- B** They cool the air in the room.
- C** They increase the rate of evaporation from the skin.
- D** They speed up the vibration of air molecules.

77. O/N/12/12/Q20

A strip is made from two metals joined together. The diagrams show the strip at room temperature and after it has been cooled.



at room temperature



below room temperature

The change in shape occurs because

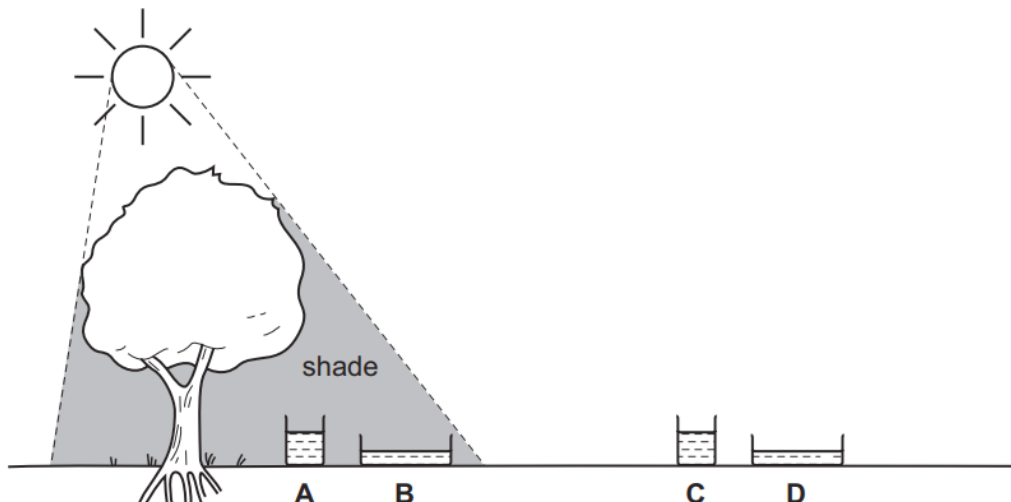
- A** brass contracts more than invar.
- B** brass expands when it cools down.
- C** invar and brass contract by equal amounts.
- D** invar contracts more than brass.

78. M/J/12/11/Q16

Four containers are filled with equal volumes of water at the same temperature.

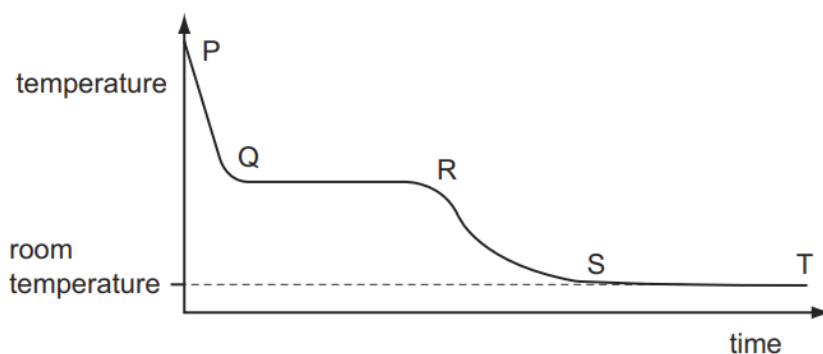
Containers **A** and **B** are shaded by a tree. Sunlight falls on containers **C** and **D**.

From which container does all the water evaporate first?



79. M/J/12/11/Q19

A hot liquid is allowed to cool. The graph shows the cooling curve.



In which part of the curve is latent heat released?

- A** PQ **B** QR **C** RS **D** ST

80. M/J/12/11/Q20

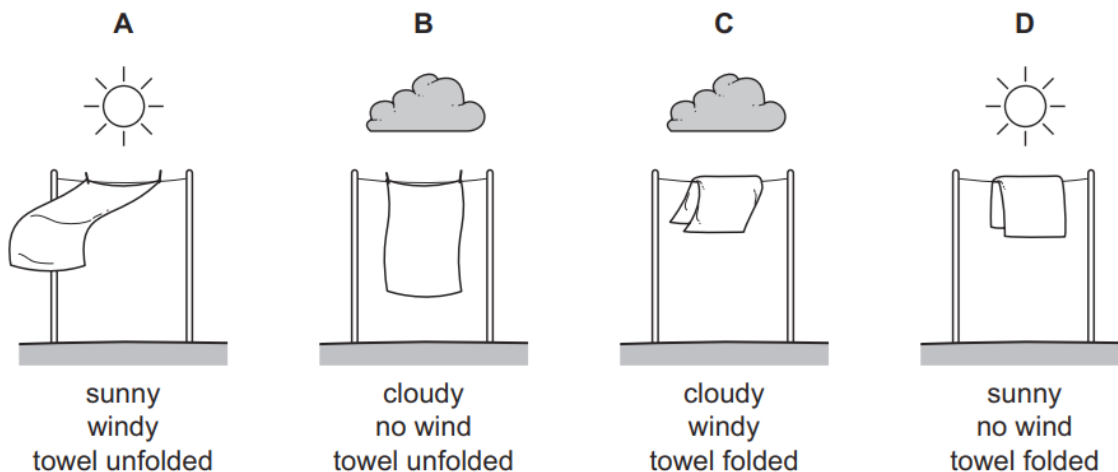
For the same temperature rise and the same original volume, which of the three states of matter expands the most and which state expands the least?

	expands most	expands least
A	gas	solid
B	liquid	gas
C	solid	gas
D	solid	liquid

81. M/J/12/12/Q16

Four wet towels are hung out to dry as shown.

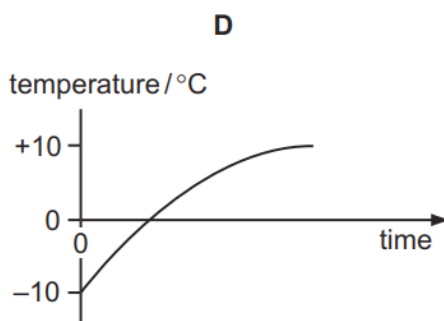
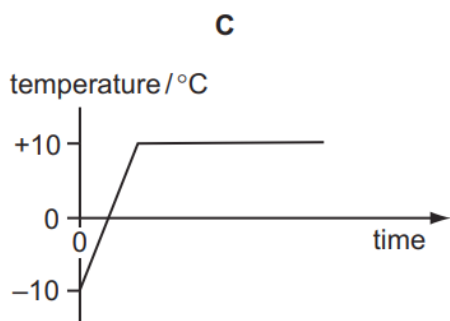
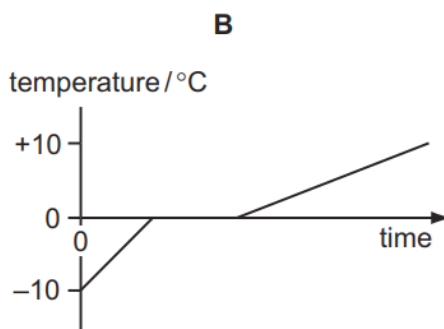
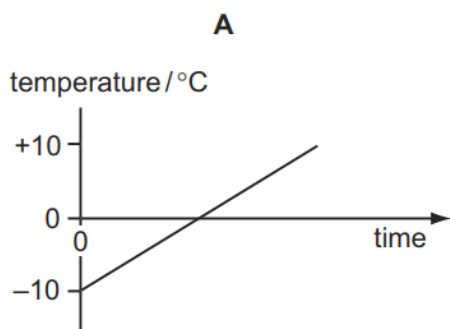
Which towel dries most quickly?



82. M/J/12/12/Q19

Ice at -10°C is heated until it is water at $+10^{\circ}\text{C}$.

Which graph shows how the temperature changes with time?



83. O/N/11/11/Q17

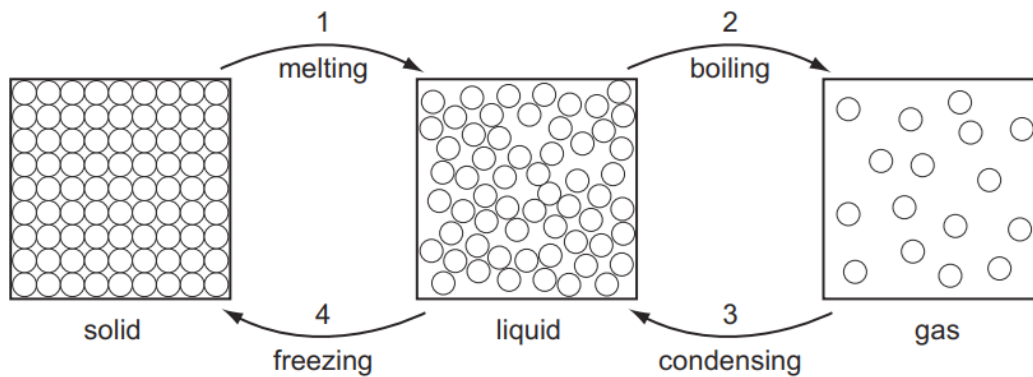
To raise the temperature of a 2.0 kg block of metal by 20°C , energy of 5.2 kJ is needed.

What is the value of the specific heat capacity of the metal?

- A** 0.13 J / (kg $^{\circ}\text{C}$)
- B** 52 J / (kg $^{\circ}\text{C}$)
- C** 130 J / (kg $^{\circ}\text{C}$)
- D** 52 000 J / (kg $^{\circ}\text{C}$)

84. O/N/11/11/Q19

Substances can change from one state to another as shown.



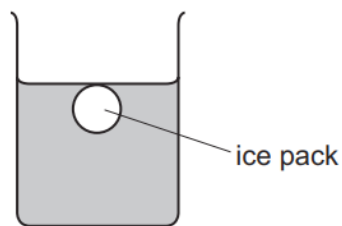
For substances to change from one state to another, there must be some energy transfer.

Which changes involve the substance taking in energy and which changes involve the substance giving out energy?

	energy taken in	energy given out
A	1 and 2	3 and 4
B	1 and 3	2 and 4
C	2 and 4	1 and 3
D	3 and 4	1 and 2

85. M/J/11/11/Q18

An ice pack is used to cool 0.25 kg of water. The specific heat capacity of water is $4.2 \text{ kJ}/(\text{kg}^\circ\text{C})$.



How much thermal energy (heat) must the ice pack extract from the water to reduce the water temperature by 15°C ?

- A** 0.070 kJ **B** 1.1 kJ **C** 16 kJ **D** 250 kJ

86. M/J/11/11/Q19

Fillings in teeth should be made from a material which

- A** does not expand when heated.
B expands by the same amount as the tooth when heated.
C expands less than the tooth when heated.
D expands more than the tooth when heated.

87. O/N/10/11/Q16

A liquid evaporates rapidly.

Why does this cause it to cool?

- A** Air molecules remove heat by contact with the liquid surface.
- B** Energy is lost by convection currents.
- C** Some of the most energetic molecules leave the liquid.
- D** The molecules have less room to move around.

88. O/N/10/11/Q20

What is caused by the thermal expansion of a substance when heated?

- A** a decrease in the resistance of a tungsten-filament lamp when switched on
- B** a rise in the pressure of the gas trapped in a gas cylinder placed in hot water
- C** the blowing of the fuse in a circuit when the current becomes too large
- D** the upward movement of the air above a Bunsen burner when it is lit

89. M/J/10/11/Q14

The liquid in a puddle evaporates and this causes its temperature to change.

How does the temperature of the liquid change and why?

	change	reason
A	decreases	less energetic molecules leave the liquid
B	decreases	more energetic molecules leave the liquid
C	increases	less energetic molecules leave the liquid
D	increases	more energetic molecules leave the liquid

90. O/N/09/Q16

A block of metal has a mass of 2.0 kg. Its specific heat capacity is $800 \text{ J/(kg } ^\circ\text{C)}$.

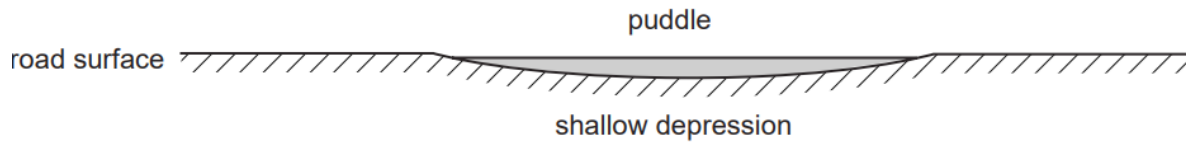
The block is supplied with 2400 J of energy.

What is the rise in temperature?

- A** 0.17°C **B** 0.67°C **C** 1.5°C **D** 6.0°C

91. O/N/09/Q18

The diagram shows a cross-section through a rain-water puddle formed in a shallow depression in a road surface.



Over a period of time, the air temperature, wind speed and wind direction all remain constant.

What happens to the rate of evaporation of water from the puddle?

- A** It decreases, because the surface area decreases.
- B** It increases, because the puddle gets shallower.
- C** It increases, because the surface area decreases.
- D** It remains constant.

92. M/J/09/Q17

What happens when a metal bar is heated?

- A** The distance between the molecules increases, making the bar longer.
- B** The molecules get larger, making the bar longer.
- C** The molecules vibrate more quickly, making the bar denser.
- D** The speed of the molecules increases, making the bar thinner.

93. O/N/08/Q20

When ice melts to become water, which force must be overcome?

- A** the attraction between electrons and the nucleus
- B** the attraction between the atoms in a molecule
- C** the force between molecules
- D** the force of gravity

94. O/N/08/Q21

Which factors increase the rate of evaporation of a liquid?

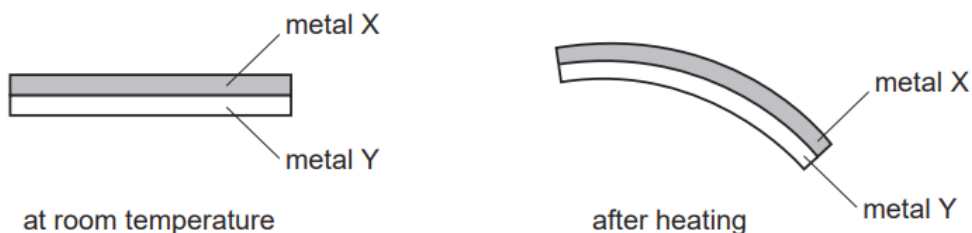
	increasing its temperature	increasing its surface area	increasing its depth
A	yes	yes	yes
B	yes	yes	no
C	yes	no	yes
D	no	yes	yes

95. O/N/08/Q19

The table shows the increase in length of four metals when heated through the same temperature rise. Each metal initially has the same length.

metal	increase in length/m
aluminium	0.000030
copper	0.000020
platinum	0.000009
steel	0.000010

A bimetallic strip is made from two of the metals. When heated, it bends in the direction shown.

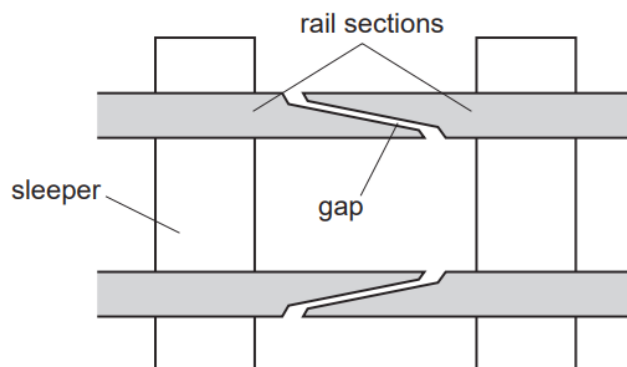


Which metals produce the above effect?

	metal X	metal Y
A	aluminium	platinum
B	copper	aluminium
C	steel	copper
D	platinum	steel

96. M/J/08/Q16

At regular intervals along a railway line there is a gap between the rail sections.



What is the reason for the gap between the rail sections?

- A** to allow for expansion of the rail sections during hot weather
- B** to allow for vibrations of the rail sections as the train passes over them
- C** to allow rain water to drain from the rail sections
- D** to keep the wheels of the train and carriages on the rail sections

97. M/J/08/Q17

Ice is taken from a freezer and left in a room. The ice melts and eventually the water reaches room temperature.

Which energy transfers take place?

	energy transfer during melting	energy transfer after melting
A	from ice to room	from water to room
B	from ice to room	from room to water
C	from room to ice	from room to water
D	from room to ice	from water to room

98. M/J/08/Q18

Some air is trapped inside a small balloon. The average kinetic energy of the air molecules in the balloon is increased.

What remains the same?

- A** the density of the air in the balloon
- B** the mass of the air in the balloon
- C** the temperature of the air in the balloon
- D** the volume of the air in the balloon

99. M/J/08/Q19

A student is investigating the evaporation of water.



The student can change:

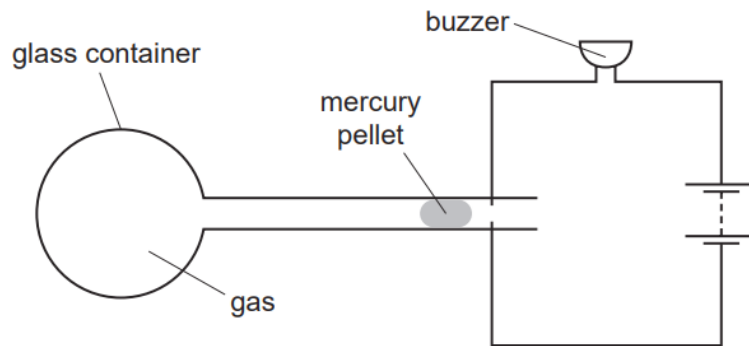
- the depth of the water;
- the surface area of the water;
- the temperature of the water.

How many of these changes, if any, would alter the rate at which evaporation occurs?

- A** 0
- B** 1
- C** 2
- D** 3

100. O/N/07/Q17

The diagram shows the design for an alarm.



The mercury pellet moves and completes the circuit.

Why does this happen?

	temperature	gas
A	falls	contracts
B	falls	expands
C	rises	contracts
D	rises	expands

101. M/J/07/Q13

A liquid is heated.

Which statement is **incorrect**?

- A** The molecules expand.
- B** The molecules gain energy.
- C** The molecules move further apart.
- D** The molecules move faster.

102. M/J/07/Q16

What is the definition of *heat capacity*?

- A** the quantity of heat required to raise the temperature of an object through 1°C
- B** the quantity of heat required to raise the temperature of 1 kg of a substance through 1°C
- C** the quantity of heat required to convert an object from solid to liquid without a change in temperature
- D** the quantity of heat required to change 1 kg of a substance from solid to liquid without a change in temperature

103. O/N/06/Q14

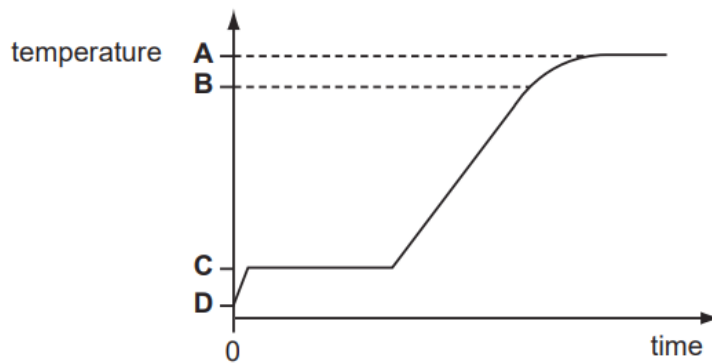
What conditions of temperature and surface area would produce the most rapid evaporation from a pool of water on a road surface?

	temperature	surface area
A	high	large
B	high	small
C	low	large
D	low	small

104. M/J/06/Q18

Some ice cubes are taken from a deep-freeze and placed in a metal container. The container is heated at a constant rate and readings of temperature and time are taken. The results are recorded on a graph.

Which temperature corresponds to 0°C ?

**105. M/J/06/Q19**

The diagrams show a bimetallic strip when it is at room temperature and after it has been cooled.



at room temperature



below room temperature

The change in shape occurs because

- A** brass contracts more than invar.
- B** brass expands when it cools down.
- C** invar and brass contract by equal amounts.
- D** invar contracts more than brass.

106. O/N/05/Q15

What will **not** affect the rate of evaporation from the surface of a liquid?

- A depth of the liquid
- B draughts above the surface of the liquid
- C surface area of the liquid
- D temperature of the liquid

107. M/J/05/Q14

Which factors increase the rate of evaporation of a liquid?

	increasing its temperature	increasing its surface area	increasing its depth
A	yes	yes	yes
B	yes	yes	no
C	yes	no	yes
D	no	yes	yes

108. M/J/05/Q16

In an experiment to find the specific heat capacity of a metal, it is found that 5200 J is needed to raise the temperature of a 2 kg block by 20 °C.

What value for the specific heat capacity is given by these results?

- A 130 J/(kg °C)
- B 520 J/(kg °C)
- C 52 000 J/(kg °C)
- D 104 000 J/(kg °C)

109. O/N/04/Q19

The fillings for a hole in a tooth should be made from a material that

- A expands more than the hole in the tooth.
- B expands by the same amount as the hole in the tooth.
- C expands less than the hole in the tooth.
- D does not expand when heated.

110. O/N/4/Q20

Heat energy is supplied at the same rate to 100 g of paraffin and to 100 g of water in similar containers.

Why does the temperature of the paraffin rise more quickly?

- A** The paraffin has a larger specific heat capacity than water.
- B** The paraffin has a smaller specific heat capacity than water.
- C** The paraffin is less dense than water.
- D** The paraffin is more dense than water.

ANSWERS

1. D	48.D	97.C	
2. D	49.C	98.B	
3. C	50.C	99.C	
4. B	51.C	100.	D
5. A	52.B	101.	A
6. B	53.C	102.	A
7. B	54.C	103.	A
8. A	55.B	104.	C
9. D	56.C	105.	A
10.A	57.A	106.	A
11.A	58.B	107.	B
12.A	59.B	108.	A
13.B	60.D	109.	B
14.D	61.B	110.	B
15.B	62.A		
16.C	63.B		
17.C	64.D		
18.D	65.A		
19.D	66.A		
20.B	67.C		
21.A	68.D		
22.C	69.A		
23.C	70.D		
24.A	71.D		
25.C	72.A		
26.C	73.A		
27.C	74.D		
28.D	75.D		
29.D	76.C		
30.B	77.A		
31.C	78.D		
32.C	79.B		
33.D	80.A		
34.C	81.A		
35.D	82.B		
36.C	83.C		
37.D	84.A		
38.A	85.C		
39.B	86.B		
40.A	87.C		
41.D	88.D		
42.B	89.B		
43.D	90.C		
44.D	91.A		
45.B	92.A		
46.C	93.C		
47.B	94.B		
	95.A		
	96.A		

1. O/N/23/22/Q5

Fig. 5.1 shows a device that cools the air in a room by evaporation.

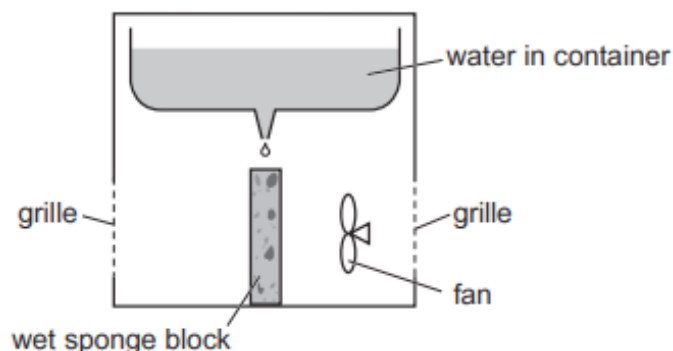


Fig. 5.1

Water drips from the container into the sponge block and keeps it wet.

Air flows in through one grille and out through the other.

As the air flows through the wet sponge block, the water in the sponge block evaporates.

(a) When the water evaporates, the temperature of water remaining in the sponge block decreases.

(i) Explain, in terms of the movement of particles, why this decrease in temperature occurs

.....

.....

.....

..... [2]

(ii) Explain why energy is needed when the water evaporates.

.....

..... [1]

(b) State **two** ways in which evaporation differs from boiling.

1

.....

2

..... [2]

[Total: 5]

2. M/J/23/22/Q4

Fig. 4.1 shows the particles (atoms) at one instant in a sample of iron at a temperature below its melting point.

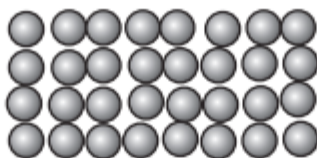


Fig. 4.1

- (a) (i) State the lowest possible temperature on the Celsius scale and on the Kelvin scale.

Celsius scale °C Kelvin scale K [1]

- (ii) The temperature of the solid increases. The sample remains a solid.

Describe how the motion of the particles changes.

.....
 [2]

- (iii) The solid melts.

State what happens to the internal energy and the temperature of the solid as it melts.

internal energy
 temperature [2]

- (b) A student:

- places a 300g piece of iron in boiling water until the iron is at a temperature of 100 °C
- removes the iron from the water and places it immediately into 100 g of water at 25 °C.

The iron cools and the water warms until both reach the same temperature, 44 °C.

The specific heat capacity of water is 4.2 J/(g °C). No energy is lost to the surroundings.

- (i) Calculate the change in energy (internal energy) of the water as it warms up.

change in energy = J [2]

- (ii) Calculate the specific heat capacity of iron.

specific heat capacity = J/(g °C) [2]

[Total: 9]

3. M/J/23/21/Q5

Fig. 5.1 shows the particles (molecules) in a sample of liquid water.

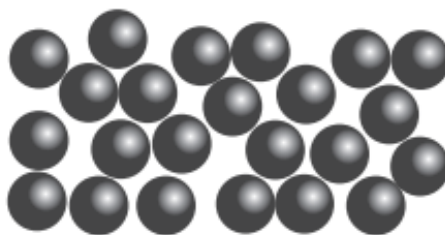


Fig. 5.1

- (a) Explain, using ideas about particles, why liquids expand more than solids for the same temperature rise.

.....
.....
.....
..... [2]

- (b) The boiling point of water is 100°C .

- (i) State the boiling point of water on the Kelvin scale of temperature.

boiling point = K [1]

- (ii) The temperature remains constant as water turns from liquid to gas at the boiling point.

Explain, in terms of particles, why energy must be provided even though the temperature stays constant.

.....
..... [1]

- (c) An electric heater is used to heat a sample of metal, as shown in Fig. 5.2. There is no thermal energy transferred from the metal to the surroundings during the heating.

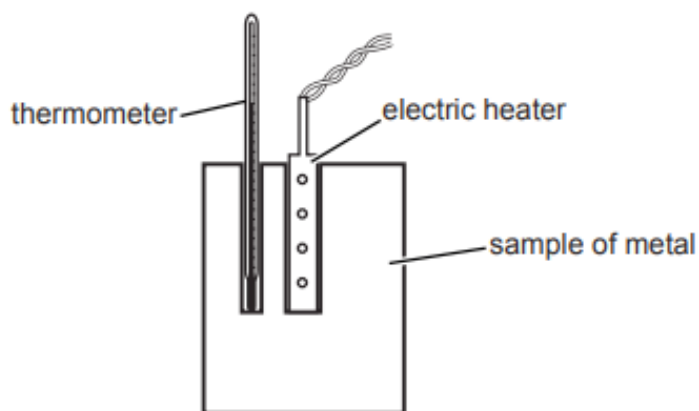


Fig. 5.2

The metal has a mass of 200g and is initially at 15 °C.

The heater is switched on for 2.0 minutes and then switched off. The maximum temperature reached by the metal is 40 °C.

The power of the heater is 20 W.

Calculate the specific heat capacity of the metal.

Show your working.

specific heat capacity = J/(g °C) [3]

[Total: 7]

4. M/J/21/22/Q9

(a) Fig. 9.1 represents the conversion of energy in a mobile phone.



Fig. 9.1

(i) State the form of energy stored in the battery.

..... [1]

(ii) State the useful form of output energy from the battery.

..... [1]

(iii) State **two** useful forms of output energy from the mobile phone.

..... [2]

(b) When the battery is in use, the average current is 1.3 mA.

Determine the charge that passes through the battery in a time of 2.0 minutes.

charge = [2]

- (c) A student estimates the thermal energy produced in the battery when the mobile phone is used and uses this value to estimate the efficiency of the battery.

He measures a temperature rise of 5.0°C within the battery in a period of time when the useful energy output from the battery is 5200 J.

The mass of the battery is 110 g and its specific heat capacity is $830\text{ J/(kg }^{\circ}\text{C)}$.

- (i) Calculate the thermal energy (heat) needed to raise the temperature of the battery by 5.0°C .

thermal energy = [3]

- (ii) State what is meant by *efficiency*.

.....
..... [2]

- (iii) Calculate the efficiency of the battery.

efficiency = [2]

- (iv) Suggest and explain **one** reason why the value for the efficiency calculated in (iii) is larger than the actual efficiency of the battery.

.....
.....
.....
..... [2]

[Total: 15]

5. M/J/21/21/Q3

A hot steel rod is cooled by plunging it into cold water, as shown in Fig. 3.1.

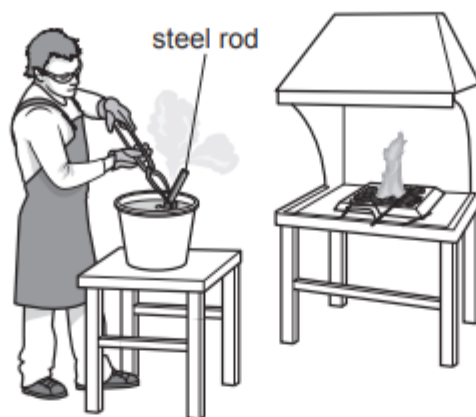


Fig. 3.1

- (a) The steel rod has a mass of 2.0 kg and is initially at a temperature of 500°C . It cools to 50°C when placed in the water.

The specific heat capacity of steel is $460\text{ J/(kg }^\circ\text{C)}$.

Calculate the thermal energy (heat) lost by the steel rod as it cools to 50°C .

thermal energy = [3]

- (b) A small mass of water boils when the rod is placed in the water. The remaining water then cools to room temperature and some of it evaporates.

In both boiling and evaporation, water molecules escape into the air.

- (i) State **one** way in which boiling is *different* from evaporation.

.....
.....
..... [1]

- (ii) The rate of evaporation decreases as the water cools.

Explain why this happens, using ideas about molecules.

.....
.....
..... [2]

6. O/N/20/21/Q9

A small glass measuring cylinder of oil is placed inside a freezer where the temperature is -18°C . Fig. 9.1 shows how the temperature of the oil varies with time t .

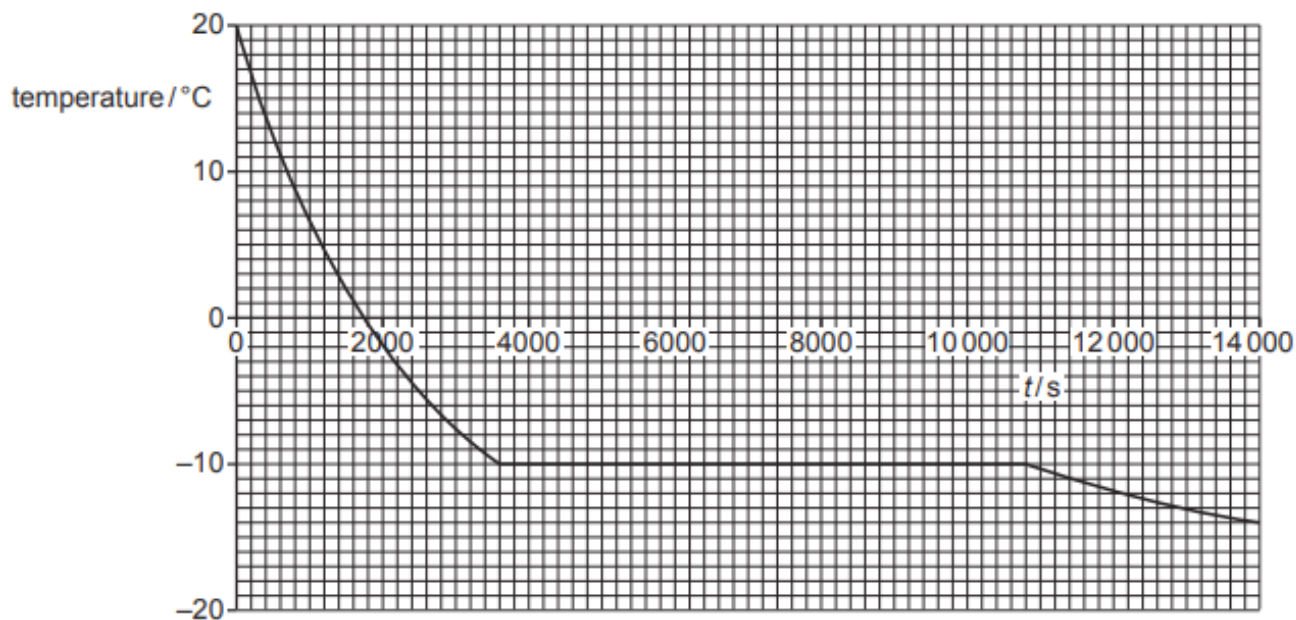


Fig. 9.1

Fig. 9.1 shows that it takes 700s for the temperature to decrease from 20°C to 10°C but that it takes 1900s to decrease from 0°C to -10°C .

(a) Suggest why these times are different.

.....

.....

.....

..... [3]

(b) Explain what happens to the molecules of the oil and what happens to the level of the oil in the glass measuring cylinder as the temperature decreases from 20°C to 0°C .

.....

.....

.....

..... [3]

- (c) (i) Determine the melting point of the oil.

..... [1]

- (ii) Explain, in terms of molecules, why the temperature of the oil does not change between $t = 3600\text{ s}$ and $t = 10\,800\text{ s}$.

.....
.....
.....
..... [3]

- (d) There is 45g of oil in the glass measuring cylinder and the specific latent heat of fusion (melting) of the oil is $5.7 \times 10^4\text{ J/kg}$.

Calculate:

- (i) the energy transferred from the oil between $t = 3600\text{ s}$ and $t = 10\,800\text{ s}$

energy = [2]

- (ii) the average rate at which energy is transferred from the oil between $t = 3600\text{ s}$ and $t = 10\,800\text{ s}$.

rate = [1]

- (e) The graph in Fig. 9.1 is steeper before the horizontal section than it is after.

Use this observation to compare the specific heat capacity of oil in the liquid and solid states and explain your reasoning.

.....
.....
.....
..... [2]

[Total: 15]

7. O/N/20/22/Q4

(a) Describe, in terms of molecules, the structure of a solid.

.....

.....

.....

..... [3]

(b) Fig. 4.1 shows a metal-worker heating a horseshoe to a high temperature before shaping it.

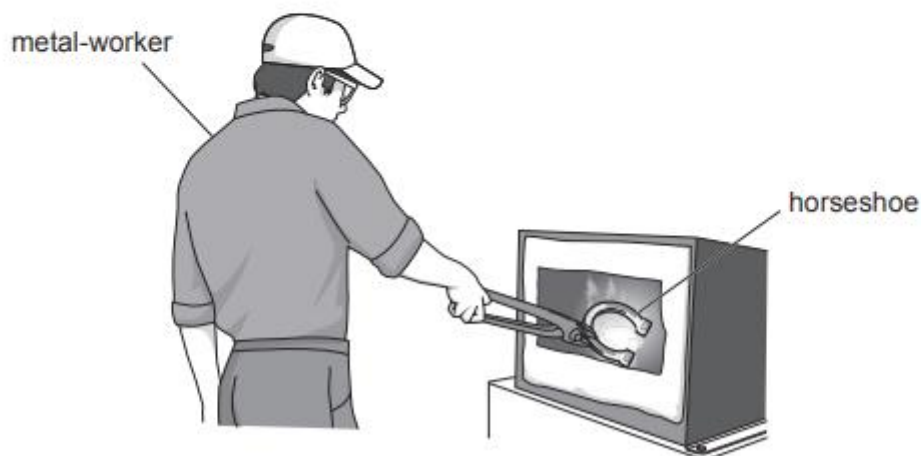


Fig. 4.1

The metal-worker then cools the horseshoe by dropping it into 8.0 kg of water at 18 °C. The final temperature of the water is 43 °C.

The specific heat capacity of water is 4200 J/(kg °C).

(i) Calculate the thermal energy transferred to the water as the horseshoe cools.

energy = [2]

(ii) The heat capacity of the horseshoe is 850 J/°C.

Calculate the temperature of the horseshoe immediately before it is dropped into the water.

temperature = [3]

8. O/N/19/21/Q3

A saucepan of water is placed on an electric hotplate and the water is brought to the boil.

As the water boils, it changes from a liquid into a gas.

(a) Describe, in terms of the molecules, **three** ways in which a gas differs from a liquid.

1.
.....
2.
.....
3.
.....

[3]

(b) In order to keep the liquid boiling, it is necessary to keep the hotplate switched on.

Explain why energy must be supplied in order to turn a liquid at its boiling point into a gas.

.....
.....
.....
.....

[2]

[Total: 5]

9. O/N/19/22/Q4

In many countries, solid salt is produced by trapping sea-water in large, shallow ponds and letting the water evaporate.

Fig. 4.1 shows salt being produced in this way.



Fig. 4.1

(a) State two ways in which evaporation differs from boiling.

1.
.....

2.
.....

[2]

(b) Describe, in terms of water molecules, what is happening as the water evaporates.

.....
.....
..... [2]

(c) (i) State why the ponds used in this process have large surface areas.

.....
..... [1]

(ii) State why this method of salt production does not work well in a country with a cold climate.

.....
..... [1]

[Total: 6]

10. M/J/18/21/Q10

(a) (i) Describe two differences between boiling and evaporation.

1.

.....

.....

2.

.....

.....

[4]

(ii) Explain, using ideas about molecules, why thermal energy is needed to boil a liquid.

.....

.....

.....

.....[2]

(b) In one type of bathroom shower, cold water passes through a metal pipe which contains an electric heater.

The cold water is heated and emerges from the shower head.

The temperature of the cold water before heating is measured and the hot water emerging from the shower in 1.0 minute is collected in a container.

Measurements and other data are:

temperature of water before heating = 16°C

temperature of water after heating = 37°C

volume of water collected in 1.0 minute = $4.6 \times 10^{-3} \text{ m}^3$

specific heat capacity of water = $4200 \text{ J/(kg }^{\circ}\text{C)}$

density of water = 1000 kg/m^3

(i) Calculate the mass of water leaving the shower in 1.0 s.

mass =[2]

- (ii) Calculate the thermal energy (heat) gained by the water in 1.0 s.

energy =[3]

11. O/N/17/21/Q3

Unwanted gold jewellery is heated in a furnace until it melts. Fig. 3.1 shows the liquid gold being poured into a mould to make a gold bar.

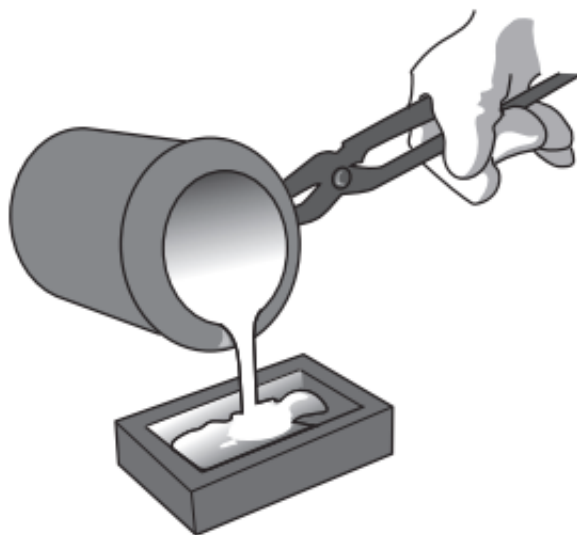


Fig. 3.1

The melting point of gold is 1100°C .

- (a)** State two ways in which the molecular structure of gold at 1200°C differs from its structure at 1000°C .

1.

.....

2.

.....

[2]

- (b)** In the furnace, 0.84 kg of solid gold reaches its melting point.

The furnace continues to supply thermal energy and, after a short time, all the gold melts.

- (i)** Explain, in terms of the molecules, why thermal energy is needed to melt the gold.

.....

.....

.....

..... [2]

- (ii)** The specific latent heat of fusion of gold is 64 kJ/kg.

Calculate the thermal energy supplied to melt all the gold.

thermal energy = [2]

12. O/N/17/22/Q10

A boiler containing water is plugged into a 230 V mains supply. The heater in the boiler is operating at full power.

(a) The power used by the heater is 2.8 kW.

(i) Calculate the current in the heater.

current =[3]

(ii) The wiring in the cable from the socket and in the boiler remains safe and does not overheat provided that the current is less than 20 A.

Suggest a suitable fuse rating for the circuit.

.....[1]

(b) The specific heat capacity of water is $4200 \text{ J/(kg } ^\circ\text{C)}$. The heater is used to increase the temperature of 6.3 kg of water.

(i) In a certain time, the temperature of the water increases from 23°C to 49°C .

Calculate the change in the internal energy of the water.

change in internal energy =[2]

(ii) The heater supplies thermal energy to the water at a constant rate. It takes more time for the temperature of the water to increase from 49°C to 75°C than from 23°C to 49°C . In both cases, the increase in temperature is the same.

Suggest why it takes more time for the temperature to increase from 49°C to 75°C .

.....
.....
.....[1]

- (c) Fig. 10.1 shows the heater at the bottom of the water in the boiler.

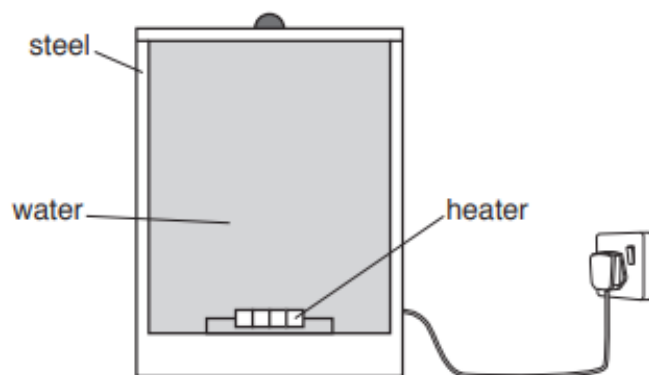


Fig. 10.1

- (i) Explain how thermal energy is transferred from the heater throughout the water.

.....
.....
.....
.....
.....[3]

- (ii) The part of the boiler that holds the water is made from steel.

Describe, in terms of particles, how thermal energy passes through the steel.

.....
.....
.....
.....
.....[3]

- (iii) State and explain, in terms of particles, what happens to the density of the steel as its temperature increases.

.....
.....
.....
.....[2]

13. M/J/17/22/Q4

Fig. 4.1 shows apparatus used to measure the specific heat capacity of a metal.

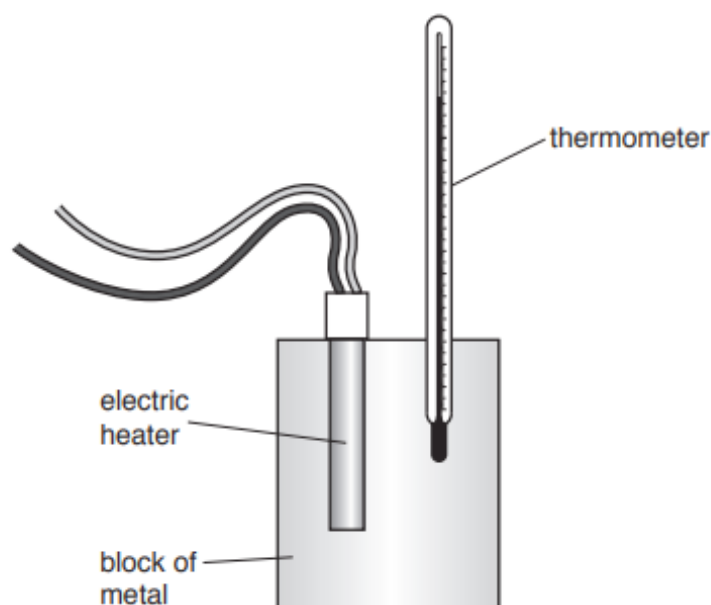


Fig. 4.1

The electric heater is switched on and supplies 17 000 J of thermal energy (heat) to the block of metal. The temperature of the metal rises from 16 °C to 38 °C. The mass of the block of metal is 850 g. No energy is lost from the metal.

(a) Calculate the specific heat capacity of the metal.

specific heat capacity =[2]

(b) Calculate the heat capacity of the block of metal.

heat capacity =[1]

(c) Explain how thermal energy (heat) is conducted through the metal.

.....
.....
.....[1]

14. O/N/16/21/Q10

A particular type of cooking oil is a liquid at room temperature.

- (a)** At room temperature, air is easily compressed but the oil cannot be compressed.

Explain, in terms of molecules,

- (i)** why the density of the oil is much greater than that of a gas,

.....
.....
.....[1]

- (ii)** why the oil cannot be compressed.

.....
.....
.....[1]

- (b)** Some of the oil is poured into a metal pan. The oil in the pan has a mass of 680g.

- (i)** The density of the oil is 0.85 g/cm^3 .

Calculate the volume of oil in the pan.

volume =[2]

- (ii)** The pan is placed on an electric hotplate, as shown in Fig. 10.1.

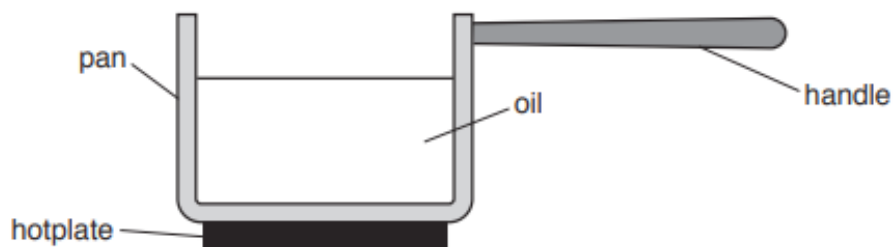


Fig. 10.1

1. Explain, in terms of molecules or free electrons, how thermal energy is conducted through the metal base of the pan.

.....

.....

.....

.....

.....

.....[3]

2. Explain how thermal energy spreads to all the oil in the pan.

.....

.....

.....

.....

.....

.....[3]

3. Suggest and explain a suitable material for the handle of the pan.

.....

.....

.....[1]

- (c) The specific heat capacity of the oil is $2.0 \text{ J/(g } ^\circ\text{C)}$. The oil is heated from 20°C to its boiling point of 260°C .

(i) State what is meant by *boiling point*.

.....
.....[1]

(ii) Calculate the thermal energy needed to heat the 680 g of oil to its boiling point.

thermal energy =[2]

(iii) Suggest one reason why the energy output from the hotplate, when heating the oil, is larger than the value calculated in (c)(ii).

.....
.....
.....[1]

15. O/N/16/22/Q4

Fig. 4.1 shows a kettle containing water placed on the burner of a gas cooker.



Fig. 4.1

The gas burner is lit at time $t = 0$. At $t = 250$ s the temperature of the water is 100°C , the boiling point of water.

(a) State what is meant by *boiling point*.

.....
.....[1]

(b) Fig. 4.2 shows how the temperature of the water changes with time t .

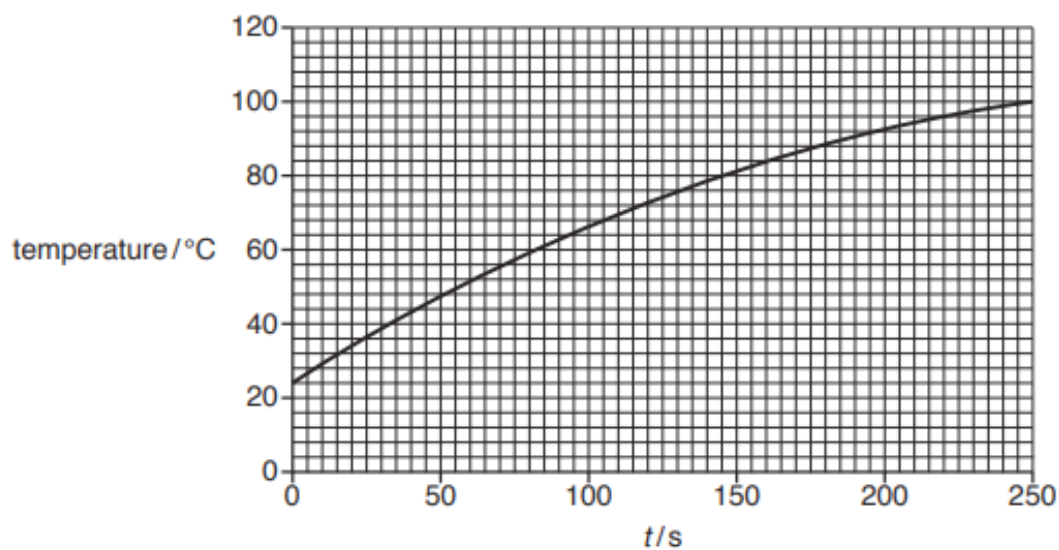


Fig. 4.2

- (i) The kettle contains 1.5 kg of water which has a specific heat capacity of $4200 \text{ J/(kg } ^\circ\text{C)}$.

Using Fig. 4.2, determine the increase in the internal energy of the water between $t = 0$ and $t = 250 \text{ s}$.

increase in internal energy =[3]

- (ii) Thermal energy (heat) is transferred to the water at a constant rate but the temperature of the water increases at a rate that is not constant, as shown in Fig. 4.2.

Explain why the temperature increases in this way.

.....
.....
.....
.....
.....[2]

- (c) When the temperature reaches 100°C , the kettle is left on the burner. Thermal energy is still supplied to the water. The water boils as the molecules form bubbles and rise to the surface.

- (i) State what happens to the temperature of the water.

.....
.....[1]

- (ii) Explain, in terms of the molecules, why it is necessary to supply thermal energy in order to keep the water boiling.

.....
.....
.....
.....
.....[2]

16. M/J/16/21/Q10

(a) (i) State what is meant by *melting point*.

.....
.....[2]

(ii) The temperature of a liquid is between its melting point and its boiling point.

Using ideas about molecules, describe the changes that occur when the liquid evaporates.

.....
.....
.....
.....
.....[3]

(b) The boiling point of nitrogen is -196°C .

Liquid nitrogen, below its boiling point, is stored in a vacuum flask, as shown in Fig. 10.1.

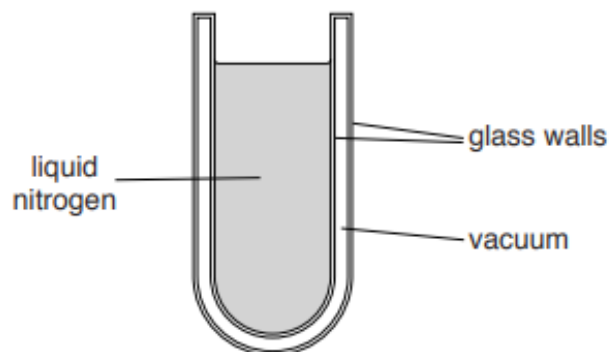


Fig. 10.1

The flask has two glass walls with a vacuum between them.

(i) Suggest where, in the apparatus shown in Fig. 10.1, evaporation occurs.

.....[1]

(ii) State and explain the advantage of having a vacuum between the two glass walls.

.....

.....

.....

.....

.....[3]

17. O/N/15/21/Q5

In a steam cleaner, water is heated to its boiling point and is then turned into steam (water in the gas state).

(a) Define *boiling point*.

.....
..... [1]

(b) (i) Describe the molecular structure of a liquid.

.....
.....
.....
.....
..... [2]

(ii) Explain, using ideas about molecules, why thermal energy (heat) must be supplied to the water as it boils.

.....
.....
.....
.....
..... [2]

18. M/J/15/22/Q4

A metal can and a plastic bottle, both containing liquid, are cooled by placing them in a jug of melting ice, as shown in Fig. 4.1.



Fig. 4.1

The can and bottle each contain 330 g of the same liquid at 15 °C.

- (a) The specific heat capacity of the liquid is 4.2 J/(g °C).

Calculate the thermal energy released when 330 g of the liquid at 15 °C cools to 2 °C.

energy = [2]

- (b) When water at 0 °C is used in the jug, instead of the melting ice, the cooling is slower.

Explain why cooling is faster when using melting ice in the jug, rather than water at 0 °C.

.....
.....
.....
..... [2]

- (c) The liquid in the metal can cools down faster than the liquid in the plastic bottle.

Suggest why this happens.

.....
.....
..... [1]

19. M/J/15/21/Q10

(a) A student makes a diagram of the molecules in a liquid, as shown in Fig. 10.1.

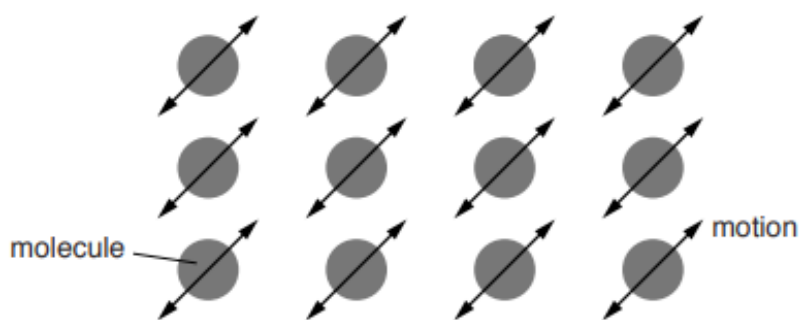


Fig. 10.1

Suggest two ways in which the student's diagram does not show the positions and motion of the molecules accurately.

1.
.....
2.
.....[2]

(b) Some of the molecules of a liquid evaporate.

(i) Describe what happens during evaporation.

.....
.....[1]

(ii) Explain why evaporation has a cooling effect on the liquid.

.....
.....
.....
.....[2]

- (c) The beaker shown in Fig. 10.2 contains pure water at a temperature of 80°C . A thermometer, initially at room temperature, is placed in the air just above the water. The reading on the thermometer starts to increase slowly.

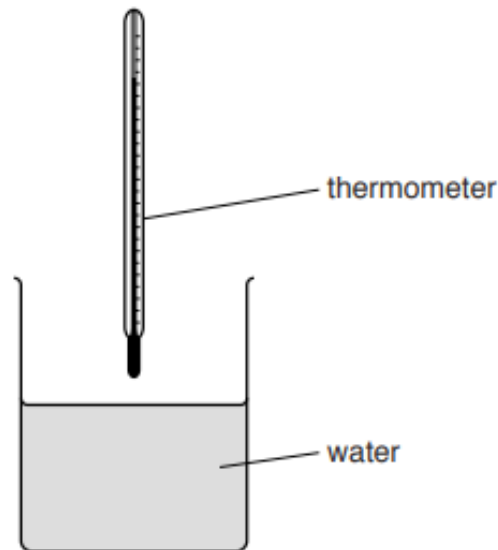


Fig. 10.2

- (i) Explain how convection in the air causes the reading on the thermometer to increase.

.....
.....[1]

- (ii) A heater is placed underneath the beaker and the water boils vigorously to produce steam. The thermometer reading increases quickly.

Explain how the water in the gas state gives thermal energy (heat) to the thermometer.

.....
.....
.....
.....[2]

20. O/N/14/22/Q6

- (a) State what happens to the molecules of a gas as its temperature increases.

.....
.....[1]

- (b) Fig. 6.1 shows a runner in a long distance race. He tips water over himself to keep cool.



Fig. 6.1

- (i) Explain, in terms of molecules, how the evaporation of water keeps the runner cool.

.....
.....
.....
.....[3]

- (ii) At one point in the race, a strong breeze blows past the runner and the water evaporates more quickly.

Suggest one reason why the water evaporates more quickly in the breeze.

.....
.....[1]

21. M/J/14/22/Q3

Most substances expand when they are heated.

- (a) (i)** State one example where expansion is useful.

..... [1]

- (ii)** State one example where expansion causes a problem.

..... [1]

- (b)** Explain, using ideas about molecules, why solids expand when heated.

.....
.....
.....
.....
..... [2]

- (c)** When equal volumes are heated through the same temperature rise, the expansions of solids, liquids and gases are different.

Complete each of the two sentences using one of these expressions:

much larger slightly larger much smaller slightly smaller

1. The expansion of a solid is than the expansion of a liquid.

2. The expansion of a gas is than the expansion of a liquid.
[2]

22. M/J/14/21/Q2

Fig. 2.1 shows apparatus used when determining the specific heat capacity of aluminium.

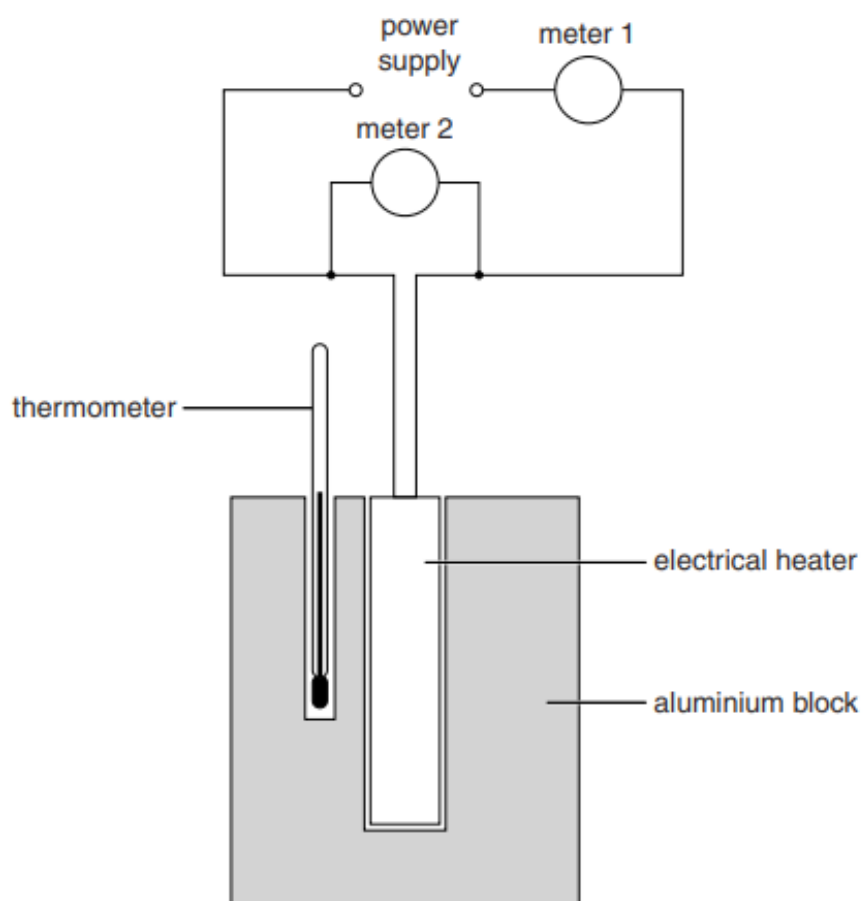


Fig. 2.1

Meter 1 and meter 2 are electrical meters.

(a) State the quantities measured by meter 1 and by meter 2.

meter 1:

meter 2:

[1]

(b) The following results are obtained in the experiment:

initial temperature of block	= 18 °C
final temperature of block	= 40 °C
mass of block	= 0.85 kg
energy supplied to heater	= 17 000 J

Assuming that no heat is lost from the block during the experiment, calculate the specific heat capacity of aluminium.

specific heat capacity = [2]

(c) In practice, some heat is lost from the block and the air above it becomes hot.

(i) Describe and explain the process by which the hot air moves away from the block.

.....
.....
.....
..... [2]

(ii) Suggest how to reduce the loss of heat from the block in the experiment.

.....
..... [1]

23. O/N/13/21/Q1

A swimming pool is filled with 450m^3 of water at 12°C .

The density of water is 1000kg/m^3 and its specific heat capacity is $4.2\text{J/(g}^\circ\text{C)}$.

(a) Calculate the mass of water in the swimming pool.

mass =[2]

(b) (i) Calculate the thermal energy (heat) needed to raise the temperature of the water to 27°C .

thermal energy =[3]

(ii) Suggest one reason why the thermal energy supplied to the water by the heating system, when raising the temperature of the water to 27°C , differs from the value calculated in **(i)**.

.....
.....[1]

24. M/J/13/22/Q11

A student places a small electrical heater inside a cup of water, as shown in Fig. 11.1.

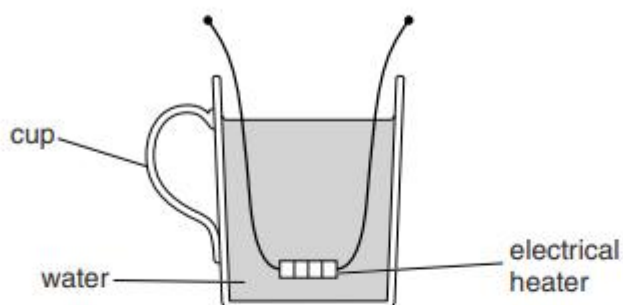


Fig. 11.1

The student determines the electrical power of the heater.

- (a) In the space above the cup on Fig. 11.1, draw the electrical circuit that the student uses. Include an ammeter, a voltmeter and a power supply. [2]
- (b) The voltage of the power supply is 12V and the current is 4.2A.
- (i) Calculate the electrical power input to the heater.

power = [2]

- (ii) Calculate the energy input to the heater in 8.0 minutes. Give your answer in kWh.

energy = kWh [3]

(c) During heating, the student notices that some of the water evaporates from the cup.

(i) Describe, using ideas about molecules of water, what happens during evaporation.

.....
.....
..... [2]

(ii) The student finds that the amount of evaporation increases when the temperature of the water is higher.

State and explain one other change that increases the amount of evaporation.

.....
.....
..... [2]

(iii) State two differences between evaporation and boiling.

1.
.....
2.
.....
..... [2]

(d) The student turns off the power supply and the water cools.

Describe and explain how convection in the air causes the water to cool.

.....
.....
.....
..... [2]

25. M/J/12/22/Q4

A beaker contains 60g of a hot substance, initially in the liquid state. Fig. 4.1 shows how the temperature of the substance changes with time t as it cools in a laboratory.

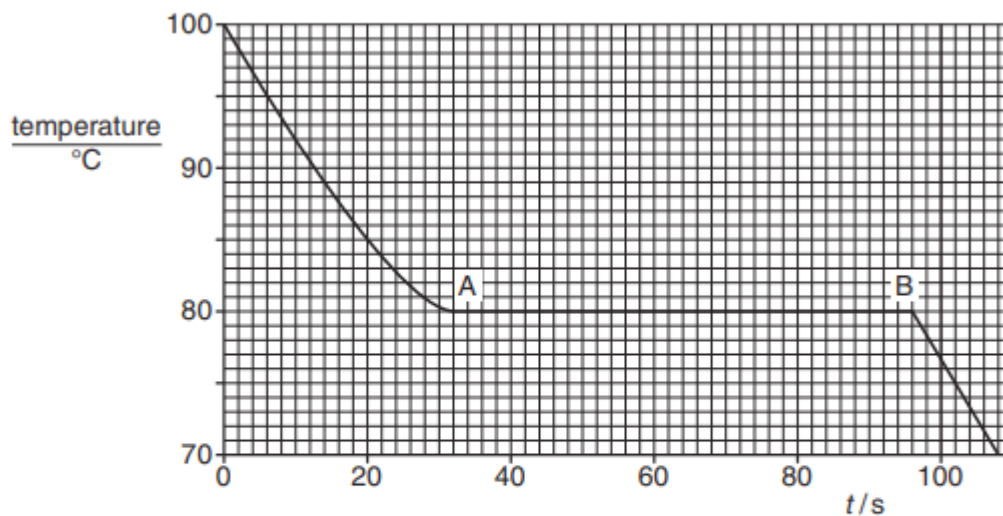


Fig. 4.1

- (a) Use Fig. 4.1 to determine the melting point of the substance.

.....[1]

- (b) The specific heat capacity of the liquid is $1.7 \text{ J/(g}^{\circ}\text{C)}$. Calculate the loss of thermal energy (heat) from the liquid between $t = 0$ and $t = 20 \text{ s}$.

loss of thermal energy =[2]

- (c) Between points A and B on Fig. 4.1, the temperature is constant as the substance changes from liquid to solid.

- (i) Explain why the temperature stays constant, even though thermal energy is lost by the substance.

.....
[2]

- (ii) Describe the change in the arrangement of the molecules as the substance changes from a liquid to a solid.

.....

 [2]

26. M/J/12/21/Q4

A student slowly heats a sample of solid wax in a test-tube.

Fig. 4.1 shows how the temperature of the wax varies with time t .

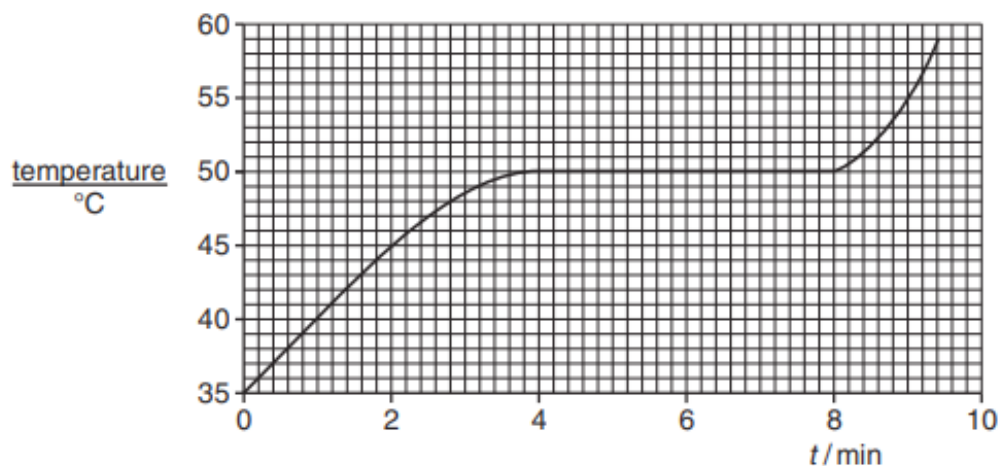


Fig. 4.1

- (a) State what is meant by the *melting point* of a substance.

.....
[1]

- (b) Thermal energy (heat) passes into the wax throughout the experiment.

- (i) Describe what effect this energy has on the wax between $t = 0$ and $t = 4$ min.

.....
[1]

- (ii) Describe and explain what effect this energy has on the wax between $t = 4$ min and $t = 8$ min.

.....

[2]

- (c) The wax expands as the temperature rises. It expands more between $t = 8$ min and 9 min than between $t = 0$ and 1 min. Explain why.

.....
[1]

27. M/J/11/22/Q3

(a) Explain, using ideas about molecules,

(i) why a balloon filled with gas expands when heated,

.....
.....
.....
.....[2]

(ii) why a balloon filled with water expands very little when heated.

.....
.....[1]

(b) (i) A bubble of gas rises from the bottom of a lake to the surface. The pressure at the bottom of the lake is $3.0 \times 10^5 \text{ Pa}$ and the pressure at the surface is $1.0 \times 10^5 \text{ Pa}$. The volume of the bubble at the bottom of the lake is 2.0 cm^3 .

Calculate the volume of the bubble at the surface.

volume =[2]

(ii) State one assumption that you have made in your calculation in (i).

.....
.....[1]

28. M/J/11/22/Q11

- (a) In an experiment to measure the specific heat capacity of water, an electric heater heats water in a glass beaker. The temperature of the water is measured at regular intervals of time. Fig. 11.1 shows how the temperature varies with time t .

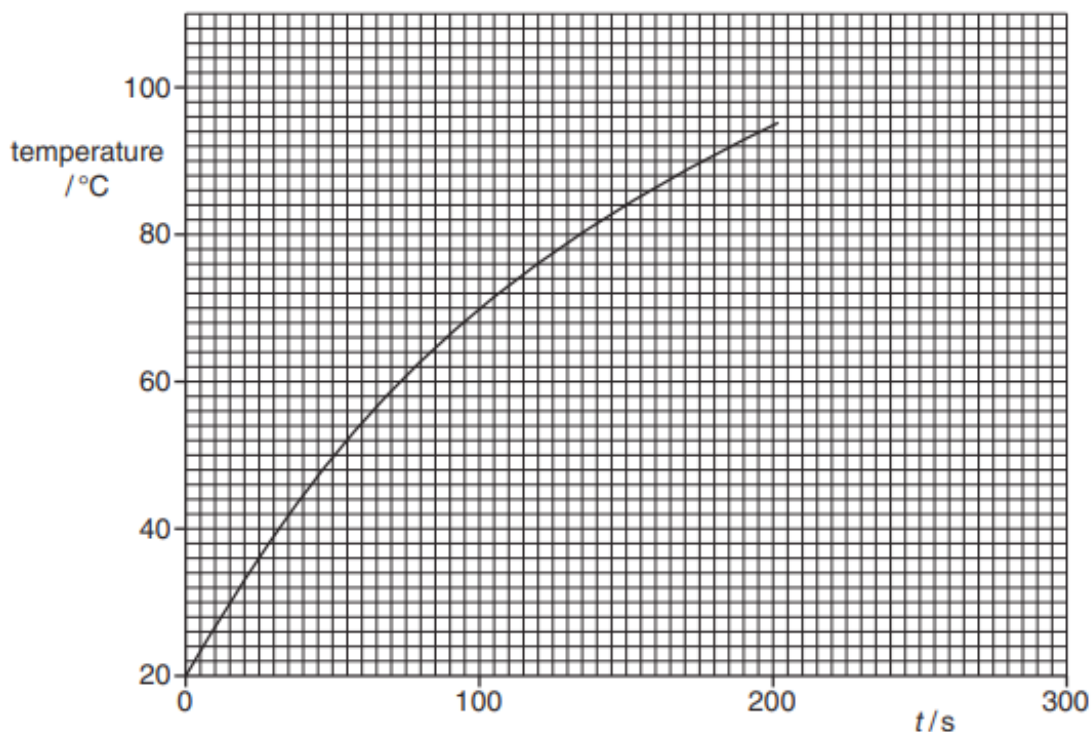


Fig. 11.1

- (i) Use Fig. 11.1 to determine the change in temperature between

$t = 0$ and $t = 100$ s,

change =

$t = 100$ s and $t = 200$ s.

change =

[1]

- (ii) State and explain why the values in (i) are different.

.....

 [2]

- (iii) Describe and explain what happens to the water if the heating is continued.

.....

 [2]

- (b) (i)** The experiment in **(a)** is repeated using 72g of water. The heater supplies 7400J of thermal energy (heat) to the water and the temperature rise of the water is 23°C.

Calculate the specific heat capacity of water.

specific heat capacity =[2]

- (ii)** A bullet of mass 72g is fired from a gun at a speed of 450 m/s.

Calculate the kinetic energy of the bullet.

energy =[3]

- (iii)** The amount of internal energy gained by the water and the amount of kinetic energy gained by the bullet are approximately equal.

Describe the change in the motion of the molecules of the water and of the molecules of the bullet that this addition of energy has caused.

water:

.....

bullet:

.....

[3]

29. M/J/10/22/Q4

A student notices puddles of water on a road, as shown in Fig. 4.1.



Fig. 4.1

Later in the day, he passes the puddles again and some of the water has evaporated.

(a) State two changes to the atmospheric conditions that would cause the water to evaporate faster.

1.

2.

[2]

(b) Explain, in terms of molecules, what happens during evaporation.

.....

.....

.....

.....

..... [2]

30. O/N/08/Q3

A river flows over a cliff, producing a waterfall. The water, at a temperature of 7.2°C , falls 700m into a pool. The gravitational field strength is 10 N/kg .

- (a) Calculate the change in the gravitational potential energy of each kilogram of water due to its fall.

energy change = [2]

- (b) Assume that all of this energy is converted into thermal energy (heat) in the water in the pool. Calculate the temperature of this water when it is in the pool. The specific heat capacity of water is $4200\text{ J/(kg }^{\circ}\text{C)}$.

temperature = [3]

31. O/N/07/Q5

Fig. 5.1 shows the arrangement of atoms in a solid block.

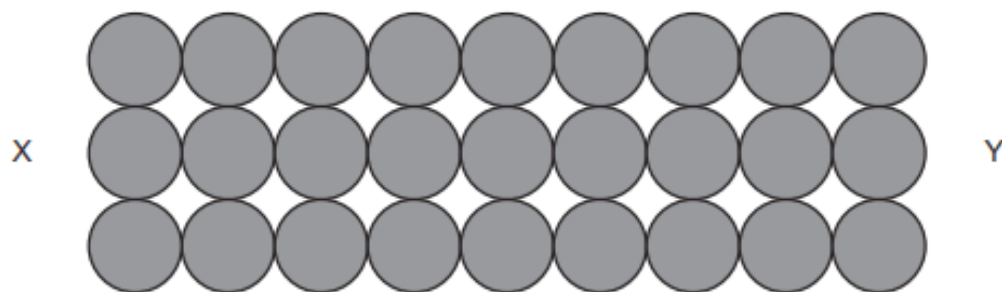


Fig. 5.1

(a) End X of the block is heated. Energy is conducted to end Y, which becomes warm.

(i) Explain how heat is conducted from X to Y by the atoms.

.....
.....
..... [2]

(ii) Explain why the solid block expands when it is heated.

.....
.....
..... [1]

(b) The block is heated and becomes a liquid.

Describe the changes that occur to the arrangement and the motion of the atoms.

.....
.....
.....
..... [2]

32. M/J/04/Q3

Thermal expansion can be a problem.



gap in
bridge

Fig. 3.1

Small gaps are left in railway lines and in bridges, as shown in Fig. 3.1.

(a) Explain in detail why the gaps are needed.

.....

.....

.....

.....

.....[3]

(b) State one other problem caused by thermal expansion and explain how it can be solved.

.....

.....

.....[1]

ANSWERS

1.

Question	Answer	Marks
5(a)(i)	faster-moving <u>particles</u> and (more likely to) escape (from the water)	B1
	remaining <u>particles</u> have less (average) kinetic energy / move more slowly (on average)	B1
5(a)(ii)	(work is done) moving particles apart (against the forces of attraction) or (energy needed) to supply the latent heat	B1
5(b)	any two from: no bubbles formed or occurs only at the surface occurs at any temperature is affected by exposed surface area / moving air over surface / temperature	B2

2.

Question	Answer	Marks
4(a)(i)	–273 (°C) and 0 (K)	B1
4(a)(ii)	particles) move fast(er)	B1
	<u>vibrate</u> / <u>oscillate</u> with larger amplitude	B1
4(a)(iii)	internal energy increases	B1
	temperature stays the same	B1
4(b)(i)	($E = mcT$) in any form numerical or algebraic	C1
	8000 (J)	A1
4(b)(ii)	100–44 or 56	C1
	0.48 or 0.47 (J/(g °C))	A1

3.

Question	Answer	Marks
5(a)	ANY 2 from - as temperature rises / heat added particles move faster / vibrate faster / move further apart - smaller force between particles (in liquid)	B1
	regular / fixed arrangement of particles in a solid or more space between particles in a liquid / less tightly packed / particles further apart	B1
5(b)(i)	373 (K)	B1
5(b)(ii)	(energy needed) to <u>separate</u> particles / increase potential energy of particles / overcome (intermolecular) forces (of attraction)	B1
5(c)	$E = Pt$ in any form, algebraic or numerical, seen, e.g. 2400 (J)	B1
	(shc =) E / mT in any form	B1
	0.48 (J/(g °C))	B1

4.

Question	Answer	Marks
9(a)(i)	chemical	B1
9(a)(ii)	electrical	B1
9(a)(iii)	ANY 2 from - light - sound - radio / microwave / electromagnetic (waves)	B2
9(b)	($Q =$) It in any form	C1
	0.16 C	A1
9(c)(i)	($E =$) mcT numerical or algebraic in any form	C1
	$0.11 \times 830 \times 5$	C1
	460 J	A1
9(c)(ii)	energy output / energy input	C1
	useful energy output / (total) energy input	A1
9(c)(iii)	(energy input =) $460 + 5200$ or 5660 (J) seen or $5200 \div (\text{energy in } 9(c)(i) + 5200)$	C1
	92% or 0.92	A1
9(c)(iv)	(thermal) energy <u>lost</u> / heat <u>lost</u> (from battery to environment)	B1
	energy input larger than calculated / 5660 (J) or temperature rise should be larger / is smaller than it should be or thermal energy / heat produced larger than 460 J / calculated	B1

5.

Question	Answer	Marks
3(a)	($E =$) mcT in any form algebraic or numerical	C1
	$2 \times 460 \times 450$	C1
	4.1×10^5 J	A1
3(b)(i)	any one of: - boiling occurs at one temperature / boiling point or evaporation occurs at any temperature / below boiling point - boiling occurs within liquid or evaporation occurs on the surface - evaporation increased by draughts, more surface area (and boiling does not)	B1
3(b)(ii)	(average) speed / K.E. of molecules decreases (as temperature falls) or (molecules) do not have the energy to escape / break bonds	B1
	fewer fast / energetic molecules escape or (only) fastest / high KE molecules escape	B1

6.

Question	answer	Marks
9(a)	(oil) hotter initially or cooler finally	B1
	temperature difference (between oil and freezer) decreases	B1
	loses energy faster when hotter or more slowly when cooler	B1
9(b)	molecules move more slowly or less <u>kinetic</u> energy or less internal energy	B1
	level decreases and oil contracts / volume of oil decreases	B1
	oil / liquid contracts more than glass / solid or molecules move closer	B1
9(c)(i)	-10°C	B1
9(c)(ii)	(attractive intermolecular) force between molecules or liquid solidifies	B1
	(intermolecular) potential energy decreases (as molecules move closer)	B1
	thermal energy lost (to freezer) is potential energy lost or latent heat lost	B1
9(d)(i)	$(Q =) ml_f$ or $0.045 \times 5.7 \times 10^4$	C1
	$2.6 \times 10^3 \text{ J}$	A1
9(d)(ii)	0.36 J/s or 0.36 W	B1
9(e)	specific heat capacity of liquid oil is smaller (than of oil in solid state)	B1
	the temperature decreases more quickly (in liquid state)	B1

7.

Question	Answer	Marks
4(a)	(molecules in) <u>fixed</u> places / positions / arrangement / pattern	B1
	(molecules in) lattice or in an orderly / repeating / regular arrangement	B1
	(molecules) close together or tightly packed	B1
4(b)(i)	$(Q =) mc\Delta T$ or $8.0 \times 4200 \times (43 - 18)$ or $8.0 \times 4200 \times 25$	C1
	$8.4 \times 10^5 \text{ J}$ or $840\,000 \text{ J}$	A1
4(b)(ii)	$\Delta\theta = Q / C$ or $8.4 \times 10^5 / 850$ or $988 (^{\circ}\text{C})$ or $990 (^{\circ}\text{C})$ or $1013 (^{\circ}\text{C})$	C1
	$988 (^{\circ}\text{C})$ or $990 (^{\circ}\text{C})$ or $1013 (^{\circ}\text{C})$ or $43 + 8.4 \times 10^5 / 850$	C1
	$1031 ^{\circ}\text{C}$ or $1030 ^{\circ}\text{C}$ or $1033 ^{\circ}\text{C}$ or $1000 ^{\circ}\text{C}$	A1

8.

Question	Answer	Marks
3(a)	any three from: molecules / they are further apart forces between molecules / them smaller molecules / they move in straight lines (between collisions) potential energy of molecules / them greater gas molecules move freely/randomly but liquid molecules slide over each other	B3
3(b)	(molecules separated against) intermolecular forces / bonds	B1
	work is done as molecules separate or bonds broken or latent heat supplied	B1

9.

Question	Answer	Marks
4(a)	any two from: only occurs at the surface / no bubbles produced occurs at any temperature produces cooling	B2
4(b)	molecules escape (from liquid)	B1
	fast(est) molecules / molecules with great(est kinetic) energy escape or slow(est) molecules / molecules with small(est kinetic) energy remain or (average) speed of remaining molecules decreases or molecules gain (kinetic) energy and escape	B1
4(c)(i)	rate of evaporation increases (with increasing area)	B1
4(c)(ii)	rate of evaporation decreases with decreasing temperature or takes too much time	B1

10.

Question	Answer	Marks
10(a)(i)	<i>Pair 1</i> boiling occurs at one temperature / constant temperature evaporation occurs at all / any temperature	B1
	<i>Pair 2</i> boiling occurs within / throughout liquid or causes bubbles evaporation occurs at surface	B1
	<i>Pair 3</i> boiling requires a heat <u>source</u> / affected by rate of heating evaporation causes cooling / affected by wind etc.	B1
	<i>Pair 4</i> temperature remains constant during boiling evaporation temperature drops / liquid cools	B1
10(a)(ii)	(thermal energy) used to break bonds or work done against intermolecular forces	B1
	molecules move further apart or P.E. increases or (molecules) push back the atmosphere	B1

11.

Question	Answer	Marks
3(a)	any two of: irregular arrangement (of molecules) intermolecular forces weak(er) / not held as firmly together intermolecular distances greater / more spaced out move in clusters through the liquid (not just vibrations) or positions not fixed or can slide past each other	B2
3(b)(i)	work done or forces overcome	B1
	atoms pulled apart or bonds broken	B1
3(b)(ii)	(Q =)ml or 0.84×64 or $0.84 \times 64\,000$	C1
	54 kJ or 54 000 J	A1

12.

Question	Answer	Marks
10(a)(i)	$(I =) P / V$ or 2.8 / 230 or 2800 / 230	C1
	0.012 or 12	C1
	12 A	A1
10(a)(ii)	12 A < integral number of amperes < 20 A	B1
10(b)(i)	$(Q =) m\Delta Tc$ or $6.3 \times (49 - 23) \times 4200$ or $6.3 \times 26 \times 4200$	C1
	$6.9 \times 10^5 \text{ J}$	A1
10(b)(ii)	<u>more thermal</u> energy is lost (per unit time)	B1
10(c)(i)	(when heated) water expands / volume of water increases / water becomes less dense	B1
	heated water / water with more energetic particles rises	B1
	cooler / denser water sinks or circulation or convection current	B1
10(c)(ii)	two separate approaches I / II	
	I atoms / molecules / ions / particles vibrate (more violently)	B1
	collide with neighbouring atoms / molecules / ions / particles	B1
	energy / heat / vibration passed on (to neighbour)	B1
	or II particles / atoms / molecules / ions vibrate (more violently)	B1
	atoms / molecules / ions / particles strike electrons	B1
	electrons travel through transporting energy / heat	B1
10(c)(iii)	<u>particles / molecules / atoms</u> move apart (on average)	B1
	volume increases and density decreases	B1

13.

Question	Answer	Marks
4(a)	$(c =) E / mT$ or 17000 / (22 × 850)	C1
	0.91 J / (g °C)	A1
4(b)	765 – 774 J / °C	B1
4(c)	molecules colliding against molecules or movement / diffusion / collision of (free) electrons	B1

- 14.
- (a) (i) molecules / they close together **or** small gaps between molecules B1
- (ii) molecules / they exert large (repulsive) forces B1
- (b) (i) $(V =)m/\rho$ **or** $680/0.85$ C1
 800 cm^3 **or** $8.0 \times 10^{-4}\text{ m}^3$ A1
- (ii) 1 molecules vibrate molecules vibrate B1
collide with neighbours **or** collide with electrons B1
transfer vibration / energy electrons travel through metal B1
- 2 heated / hot oil expands / less dense B1
rises B1
convection current / circulation established B1
- 3 any suitable named insulator **and** it is a poor conductor B1
- (c) (i) temperature at which (liquid) vaporises / becomes gas / steam B1
- (ii) $(Q =)mc \Delta T$ **or** $680 \times 2.0 \times (260 - 20)$ **or** $680 \times 2.0 \times 240$ C1
 $3.3 \times 10^5\text{ J}$ A1
- (iii) heat supplied to pan **or** heat lost to air / surroundings B1
- 15.
- (a) **temperature** at which liquid / water turns to gas / vapour / steam B1
- (b) (i) $(T =)24\text{ }(^{\circ}\text{C})$ **or** $100 - 24$ **or** 76 C1
 $(\Delta Q =)mc\Delta T$ **or** $1.5 \times 4200 \times 76$ C1
 $4.8 \times 10^5\text{ J}$ A1
- (ii) heat is lost (to the surroundings) **or** evaporation B1
at higher temperatures heat is lost at greater rate B1
- (c) (i) stays at $100\text{ }^{\circ}\text{C}$ / constant B1
- (ii) molecules separate / are pulled apart / are far apart / break bonds /
overcome forces of attraction B1
work done separating the molecules **or** molecules gain PE B1

- 16.
- (a) (i) temperature B1
when solid turns to liquid B1
- (ii) molecules escape (surface) C1
fastest molecules / most energetic molecules A1
escape / break bonds
leaving behind slower molecules / colder molecules B1
or temperature falls
- (b) (i) at the surface / top of liquid B1
- (ii) less heat / energy enters (liquid nitrogen) / transfers B1
or less nitrogen evaporates / boils B1
reduces / stops conduction **and** convection B1
explanation of no conduction or convection, e.g. no molecules / no medium
- 17.
- (a) **temperature** at which a liquid becomes a gas B1
- (b) (i) molecules close together / touching **or** closer than in gas B1
randomly arranged **or** irregular structure B1
- (ii) to separate / increase the distance between molecules B1
work done against (intermolecular) forces **or** supply p.e. **or** break bonds B1
- 18.
- (a) $(H=) mcT$ **or** $330 \times 4.2 \times 13$ C1
18 000 J **or** 18 020 J **or** 18 018 J A1
- (b) ice takes in / needs heat / energy B1
- for latent heat
 - to melt / turn to water (at 0 °C) / change state
 - to break bonds / for molecules to gain P.E.
- water (in jug initially at 0 °C) warms up B1
or ice (and melted water in jug)
- stays at 0 °C / stays cold / stays at constant temp.
 - gives larger temperature difference (between liquid and melting ice in jug)
- (c) metal is a good conductor (of heat) B1
or metal / can has lower heat capacity
allow opposite statements for plastic, e.g. plastic is an insulator (of heat),
penalise wrong statements and Physics, e.g. liquid evaporates from can, metals
conduct temperature / convect better

- 19.
- (a) (liquid) molecules not arranged (so) regularly B1
 (liquid) molecules not vibrating / moving in same direction B1
or do not have same speed
- (b) (i) molecules / liquid escape (from surface) / break bonds B1
- (ii) fast moving / more energetic molecules evaporate / escape B1
 leaving slow molecules **or** molecules with less **kinetic** energy (on average) B1
- (c) (i) hot air rises B1
- (ii) (steam) condenses or changes to liquid (on thermometer) B1
or heat (conducted) from hot to cold
 gives out latent heat (to thermometer) B1
or explanation involving bonds being made
- 20.
- (a) (the molecules) move faster **or** have more **kinetic** energy **or** accelerate **ignore** vibrate faster B1
- (b) (i) faster / energetic molecules escape B1
 average speed decreases **or** slower molecules remain B1
- temperature depends on average KE **or** heat taken from runner B1
OR
 liquid becomes gas / vapour
 latent heat needed or bonds broken
 heat taken from runner
- (ii) water vapour blown away **or** surrounding air less humid B1
- 21.
- (a) (i) any sensible example where expansion is useful [B1]
- (ii) any sensible example where expansion causes a problem [B1]
- (b) (molecules) move fast(er) or vibrate fast(er) or have more (kinetic / potential / internal) energy [B1]
- (molecules) move apart or distance between molecules increases or vibration has larger amplitude or vibration takes up more space or bonds stretch [B1]
- (c) slightly smaller [B1]
 much larger [B1]

- 22.
- (a) current **and** voltage /p.d./e.m.f. in correct order [B1]
- (b) (c=) E/mT in symbols or numbers e.g. $17\,000 = 0.85 \times c \times 22$ [C1]
 $910 \text{ J/(kg } ^\circ\text{C)}$ [A1]
- (c) (i) (hot air) rises **or** convection mentioned [B1]
 (hot) air less dense [B1]
- (ii) lag **or** cover with insulating material **or** warmer room **or** start with colder block [B1]
- 23.
- (a) ($m =$) ρV **or** 1000×450 C1
 $4.5 \times 10^5 \text{ kg}$ A1
- (b) (i) ($Q =$) $mc\Delta T$ **or** $4.5 \times 10^5 \times 4.2 \times 15$ **or** 4200 **and** $15/(27-12)$ C1
 $4.5 \times 10^5 \times 4200 \times 15$ **or** $2.8(35) \times 10^7$ C1
 $2.8(35) \times 10^{10} \text{ J}$ A1
- (ii) thermal/internal energy/heat lost **or** gained by something specific (e.g. air/pool walls/tiles etc.) **or** heat lost by evaporation B1
- 24.
- (a) power supply with ammeter and heater in series B1
 voltmeter in parallel with heater/ power supply B1
- (b) (i) ($P =$) VI in any form C1
or 4.2×12
- $50(.4) \text{ W}$ A1
- (ii) ($E =$) Pt i.e. any power $\times$ any time e.g. $50(.4) \times 8$ C1
- $8/60$ C1
or 0.13(3) seen
or division by 1000 seen anywhere
- $0.0067(2) \text{ (kWh)}$ A1

(c) (i) molecules escape (from surface/leave water) / become gas or vapour / break bonds C1

fast(er) moving / high energy/ energetic molecules escape A1

(ii)

change	M1	explanation
wind / draught / breeze		wind knocks molecules away
or larger surface area		more chance/possibility of escape/more space to escape or more molecules come to/near/at surface
or decrease humidity / drier air		fewer molecules return/from air
or decrease atmospheric pressure		fewer air molecules to hit during escape

(iii) evaporation occurs at surface **and** boiling inside liquid/bubbles
 evaporation occurs at any temperature (accept room temperature)
and boiling occurs at boiling point/100°C/ fixed / specific temperature
 evaporation increased by draughts/higher temp/more area **and** boiling is not
OR increase in pressure stops boiling but only reduces evaporation
 any two B2

(d) water heats air (by conduction) B1
 or water loses heat/energy (to cup or air)
 or air gains heat/energy (from water)

hot / heated air / particles rise B1
 or cold air / particles sink
 or hot air is less dense
 or cold air is more dense

25.		
(a)	80 °C	B1
(b)	(Q =) mcT in any form numerical or algebraic 1530 J	C1 A1
(c)	(i) Any 2 lines from • latent heat/energy mentioned • latent heat/energy given out/lost • bonds being made/strengthened • molecules lose PE • molecules KE constant	B2
	(ii) ANY 2 lines but max 1 if no change/comparison implied molecules change from OR in liquid • random arrangement • move throughout in some form (e.g. move freely) • move or occur in clusters change to OR in solid • regular arrangement/shape or fixed position/shape • vibrate • separation (probably) close(r)	B2
26.		
(a)	temperature when solid melts; temperature when solid changes to liquid	B1
(b)	(i) temperature increases; molecules move faster/have more kinetic energy/ vibrate faster/move further apart	B1
	(ii) change of state; solid changes to liquid	B1
	latent heat provided; break bonds; molecules move apart/break free; reduce bond strength; idea of more disorder	B1
(c)	liquids expand more than solids	B1

27.		
(a)	(i) molecules have more kinetic energy/speed/velocity hit sides hard(er)/with more force // (initially) hit sides (more) often/frequently // create large(r) pressure (initially)	B1 B1
	(ii) (larger) forces between liquid molecules/(stronger) bonds	B1
(b)	(i) $P_1 V_1 = P_2 V_2$ numerical or algebraic 6(.0) cm ³	C1 A1
	(ii) temperature is constant // no gas enters/leaves // mass constant	B1
28.		
(a)	(i) 50°C and 24/25°C	B1
	(ii) heat loss or evaporation mentioned // molecules escape // liquid to vapour more heat loss or more evaporation etc. because temperature is higher	C1 A1
	(iii) temperature becomes 100°C // reaches boiling point // temperature becomes steady water boils // water turns to steam/gas // energy loss = energy gain	B1 B1
(b)	(i) ($c =$) $E/m\Delta T$ in any form, numerical or algebraic, e.g. $7400/72 \times 23$ 4.5 or 4.47 or 4.4686 J/(g °C) // 4468.6 J/(kg °C)	C1 A1
	(ii) ($E =$) $\frac{1}{2} mv^2$ algebraic only $\frac{1}{2} \cdot 0.072 \cdot 450^2$ 7300 J // 7290 J (7 290 000 (J) alone gets 2/3)	C1 C1 A1
	(iii) <i>water molecules</i> move/vibrate fast(er)/(more) vigorously <i>water molecules</i> random motion // move (more) throughout liquid/all directions // slide over each other // move in convection // hit more often // move further apart <i>bullet molecules</i> motion in one direction/away from gun/towards target/ all have same (increase in) speed	B1 B1 B1
29.		
(a)	increased/high(er) temperature/hot(ter) wind or air flow less humidity less pressure	ANY 2 lines B2
	(b) molecules/atoms/particles escape/leave (surface) or molecules become gaseous/vapour or molecules break bonds molecules with large(est) energy/high(est) speed sufficient or enough energy escape/break bonds/become gaseous or leave behind slow/less energetic molecules	C1 A1

30.
 (a) mgh or $F \times d$ or 10×700 C1
 (-)7000 J A1
- (b) $Q/E/H = mc\Delta T$ or $(\Delta T =) 7000/(1) \times 4200$ C1
 1.7 or 1.67 or 5.5 C1
 8.9 °C e.c.f. (a) A1
31.
 (a) (i) atoms vibrate/move back and forth/to and fro (accept particles/molecules) M1
 atoms hit neighbours or pass on heat/energy to neighbour (not vibrations) A1
- (ii) atoms take up more space/further apart/larger vibrations (not atoms larger) B1
- (b) atoms move throughout (liquid) or not in fixed places or arrangement irregular or B1
 broken bonds (e.g. atoms move faster) B1
 atoms move at random/further apart (e.g. fixed volume/variable container shape etc.)
32.
 (a) rise in temperature / hot / heated B1
 road / bridge / rail / metal expands or gap reduces B1
 no buckling / deformation / breaking / cracking / twisting / tilting B1
- (b) any other problem + solution
 e.g. concrete cracks – leave a gap, telephone wires sag – put them high / tight
 hot water cracks glass – use thin glass / car engines seize up – cool them
 water freezes in pipes – lag them or use antifreeze / tyres burst – let air out
 pipes bend – use flexible joints / dashboard deforms – car in shade
 wrong readings on measuring cylinder – use correct temp. B1